

Computer Vision

01 Introduction

Peng-Shuai Wang Assistant Professor
Peking University

授课老师

- 王鹏帅 助理教授

- 主要经历:

- 2009-2018, 清华大学, 本科、博士
 - 2018-2022, 微软亚洲研究院, 历任研究员、高级研究员
 - 2022-今, 北京大学 王选计算机研究所 助理教授

- 研究兴趣: 计算机图形学/视觉, 几何深度学习, 几何处理

- 邮箱: wangps@pku.edu.cn; wangps@hotmail.com

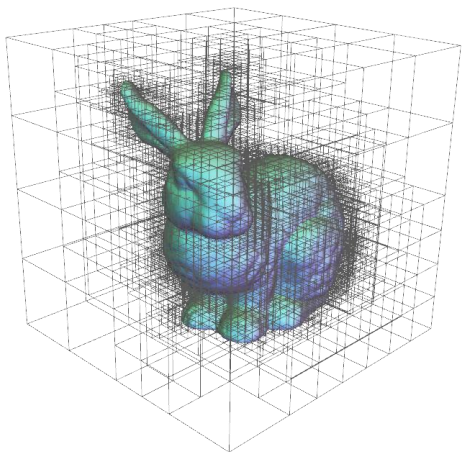
- 个人主页: <https://wang-ps.github.io>

- 代表奖项: 博雅优秀青年学者, 智源学者, AsiaGraphics 亚洲图形学青年学者奖, 中国三维视觉大会年度优秀青年学者奖, CCF-CAD/CG 青年科技奖, CVM 最佳论文奖

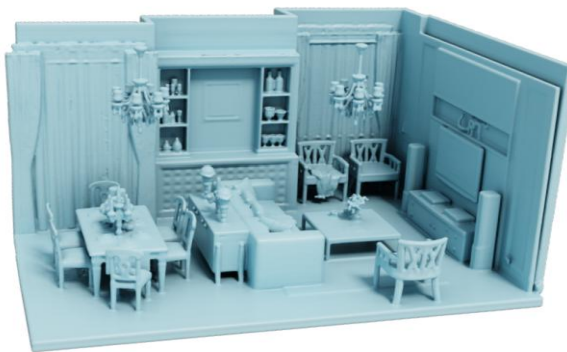
授课老师：王鹏帅 助理教授

研究方向： 计算机图形学，几何深度学习，几何处理

研究内容： 研究面向三维几何数据的通用、统一的智能图形系统，让计算机能够高效地感知、创造三维图形



O-CNN 将神经网络的运算和
存储限制在非空的三维区域
[Wang et al. SIGGRAPH 2017]



DualOGN 在500ms内可
以重建复杂三维场景
[Wang et al. SIGGRAPH 2022]



OctFormer 把注意力机制作用在
包含相同点数的不规则小窗
[Wang SIGGRAPH 2023]

课程助教



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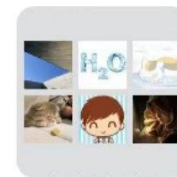


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课程信息



群聊：计算机视觉 2026

- 答疑
 - 邮件预约
 - 欢迎积极讨论关于课程内容或者研究等问题
- 上课内容
 - 英文课件 + 中文讲解
 - 课件会上传到教学网
- 微信群
 - 与各位同学建立联系；快速通知



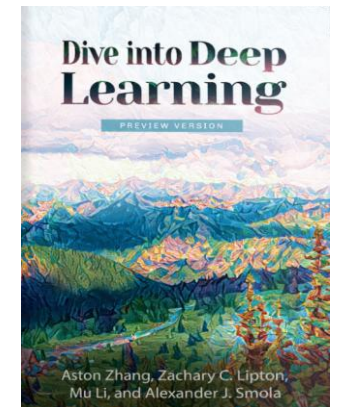
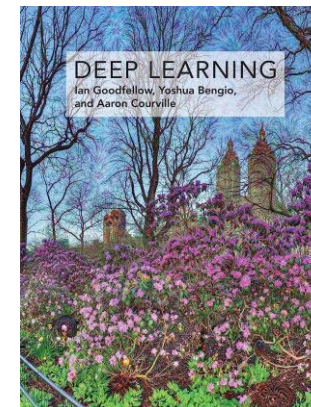
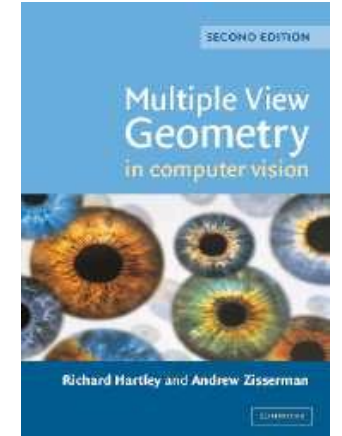
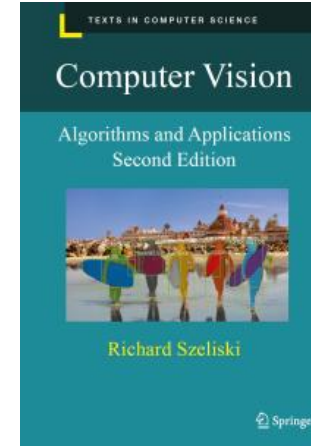
该二维码7天内(9月17日前)有效，重新进入将更新

特别鸣谢

- 本课程内容参考了以下老师的课件
 - Yixin Zhu, Siyuan Huang, Justin Johnson, Li Fei-Fei, Shree Nayar, Leonid Sigal, Andreas Geiger
 - Antonio Torralba, Bill Freeman, Phillip Isola, Rick Szeliski, Hyun Soo Park, Svetlana Lazebnik, Jan Koenderink, Matthias Niessner, Song-Chun Zhu

教材

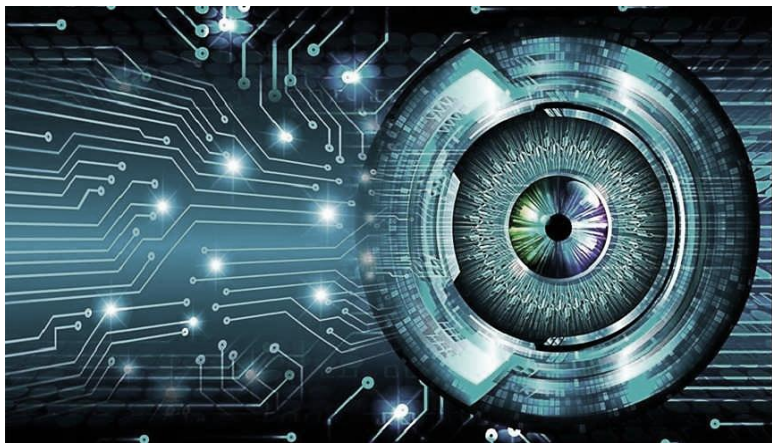
- **Computer Vision: Algorithms and Applications**
 - <https://szeliski.org/Book/>
- **Multiple View Geometry in Computer Vision**
 - <https://www.robots.ox.ac.uk/~vgg/hzbook/>
- **Deep Learning**
 - <http://www.deeplearningbook.org>
- **Dive into Deep Learning**
 - <https://d2l.ai>



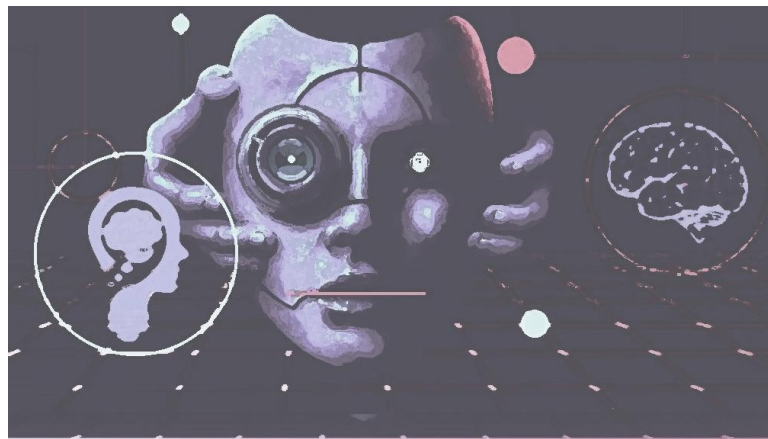
作业与考核

- 6个作业，每个作业10分，平均2周一个作业
 - 提交代码和实验报告，独立完成
 - 每迟交一天扣1分，扣一半为止
- 1个课程大作业20分
 - 最多三人一组，提交代码和报告，
 - 报告类似于学术论文，包括研究动机、相关文献、算法实现、实验结果等
- 课程小测 20 分，预计在12月份
- 课程参与，允许请假；课程参与额外奖励 2 - 4分
- 总分100+，比例会调整
- 如果发现不诚信行为，或者用了大模型但没用对，该作业记为0分

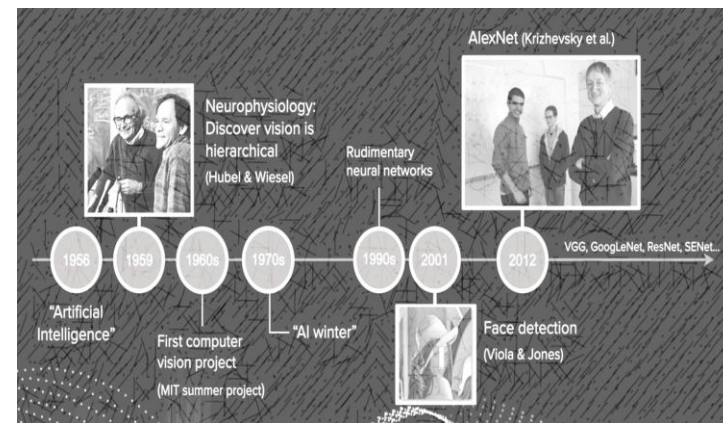
本讲内容



What is CV?

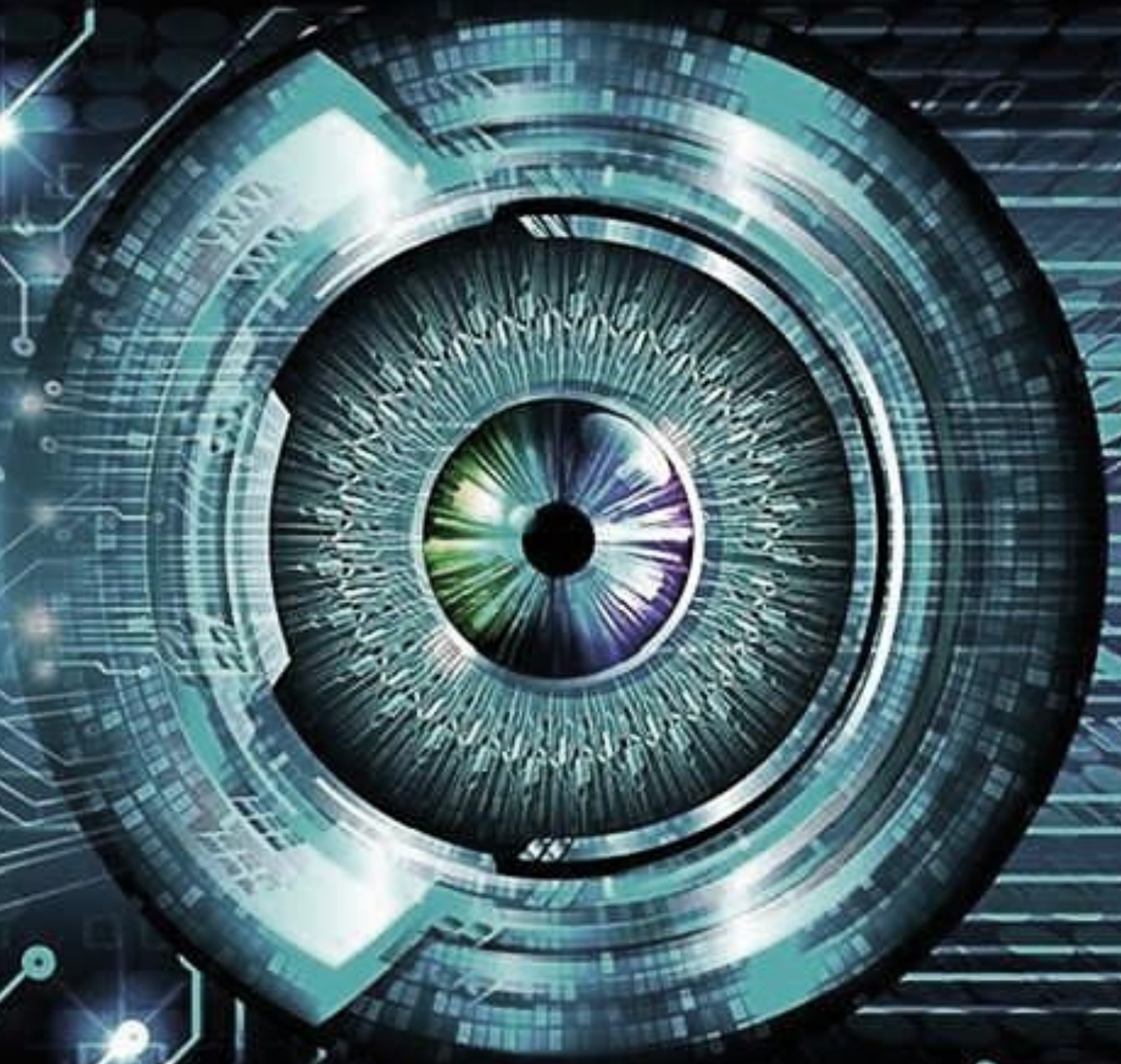


CV is Hard



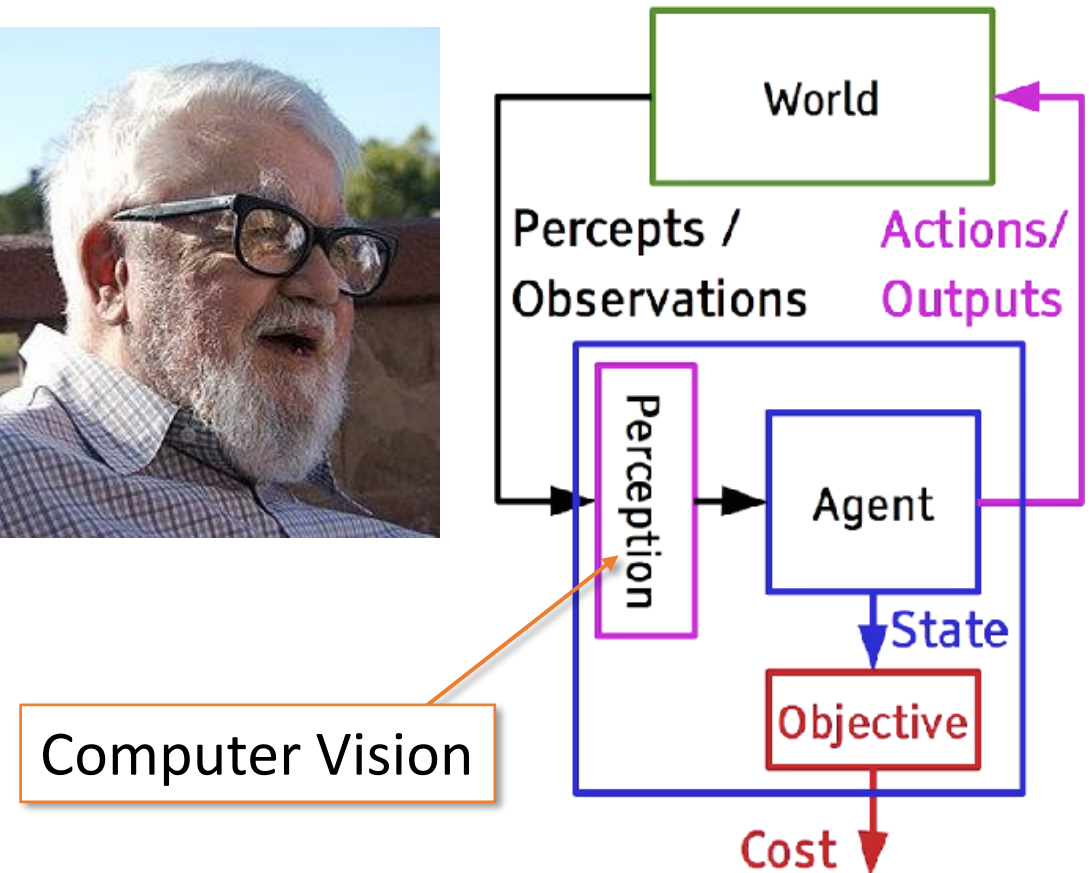
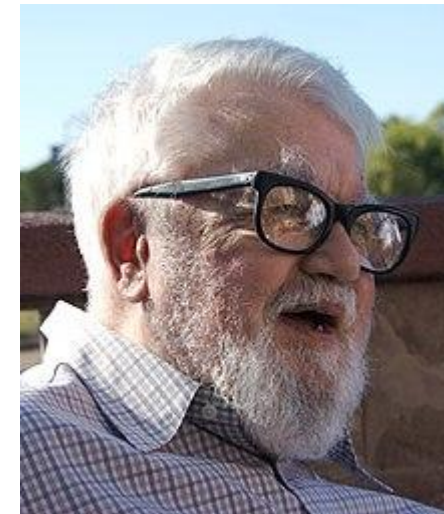
History of CV

What is CV?



Artificial Intelligence

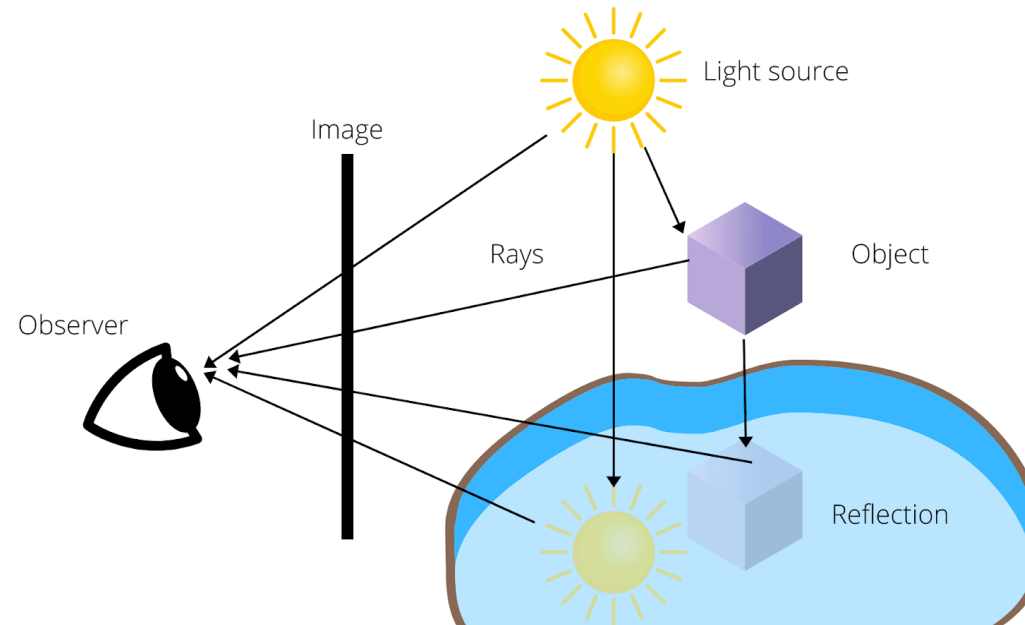
- Dr. John McCarthy
 - One of the founders of the discipline of artificial intelligence
 - The winner of 1971 Turing Award for his contributions to the topic of AI
- “An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves.”



Computer Vision

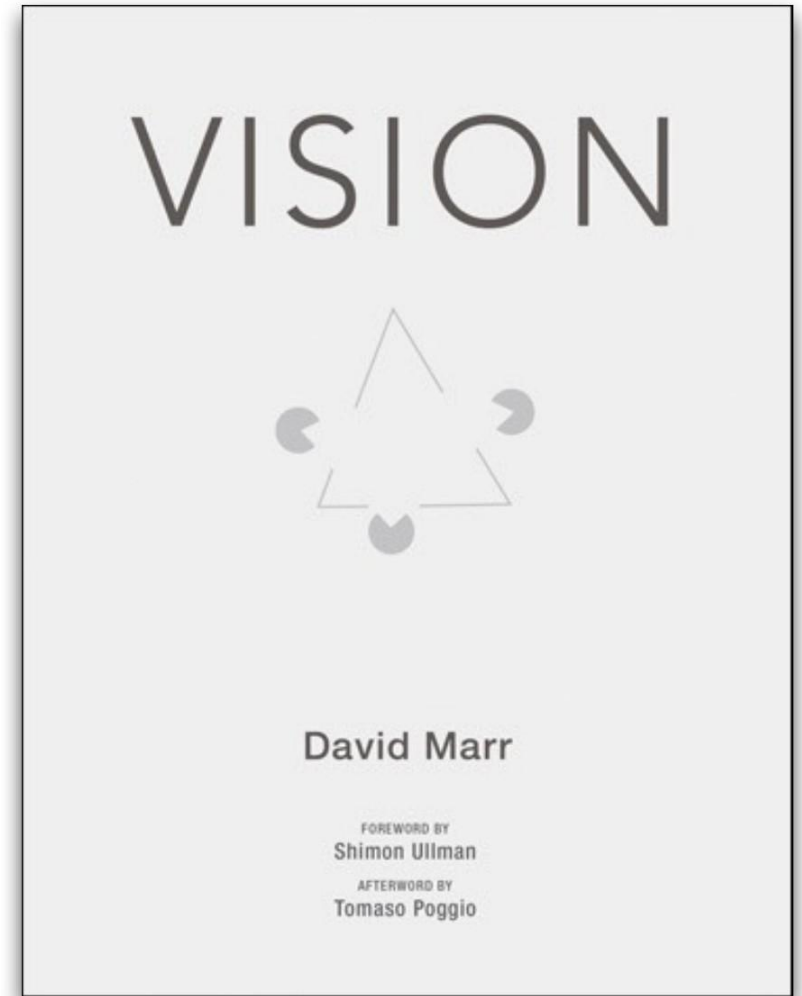
- The input of a computer vision system is sampling of the light
- The output includes **3D scene representations, semantics**, and so on

The Goal of Computer Vision is to **convert light (images) into meaning** (geometric, semantic, ...)

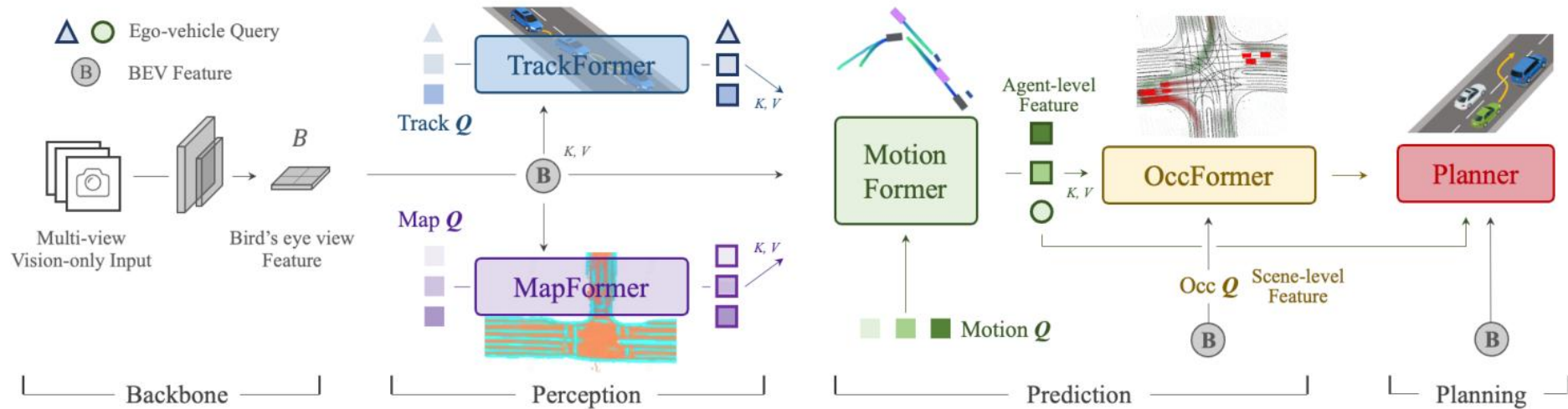
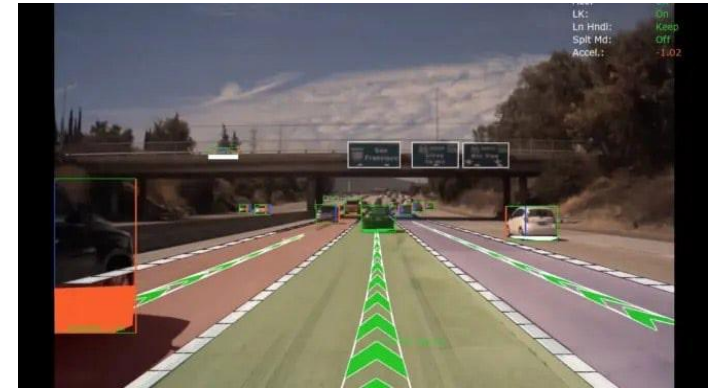
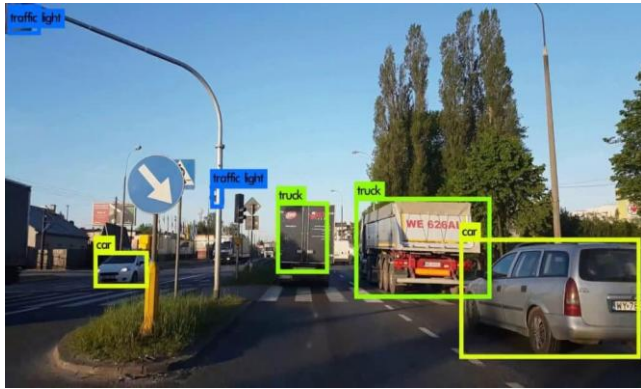


Computer Vision

- Computer vision is to make computers to see.
- What does it mean, to see? The plain man's answer (and Aristotle's, too) would be, to know what is where by looking.
- To discover from images what is present in the world, where things are, what actions are taking place, to predict and anticipate events in the world.

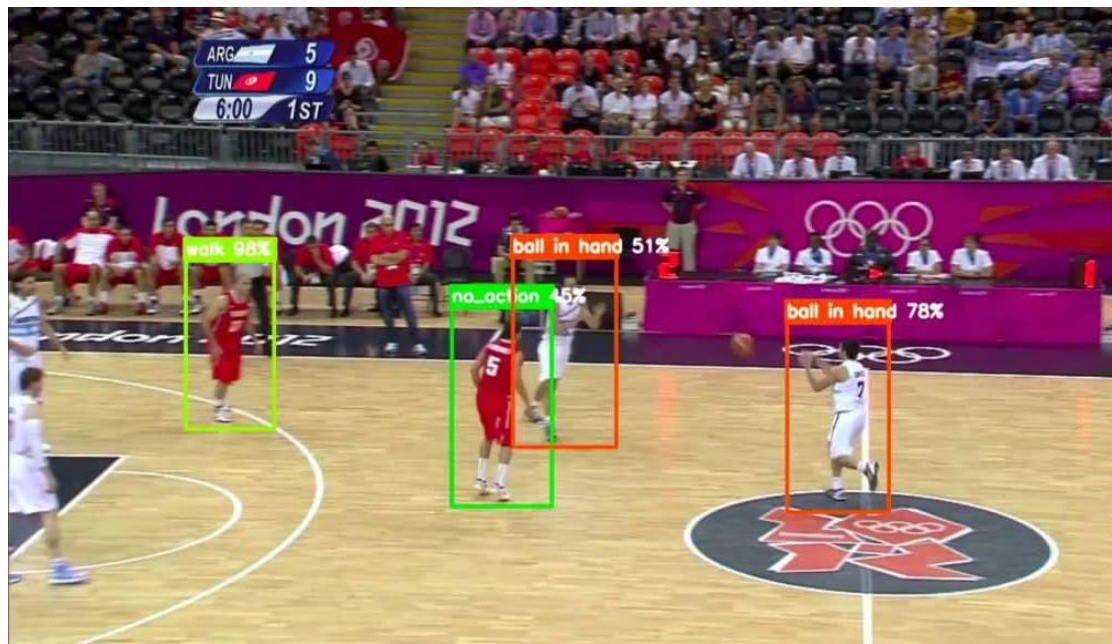


Applications: Autonomous Vehicles



CVPR 2023 Best Paper

Applications: Sports



Cut



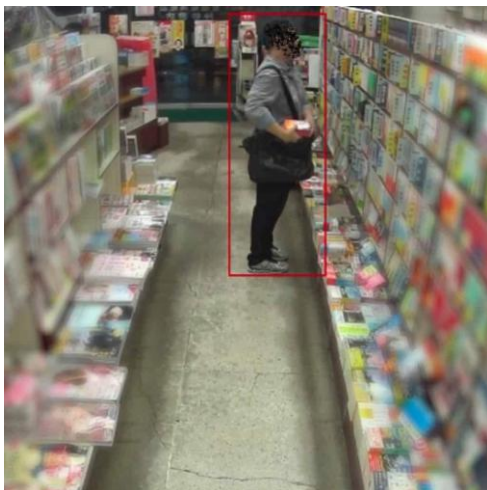
Drive



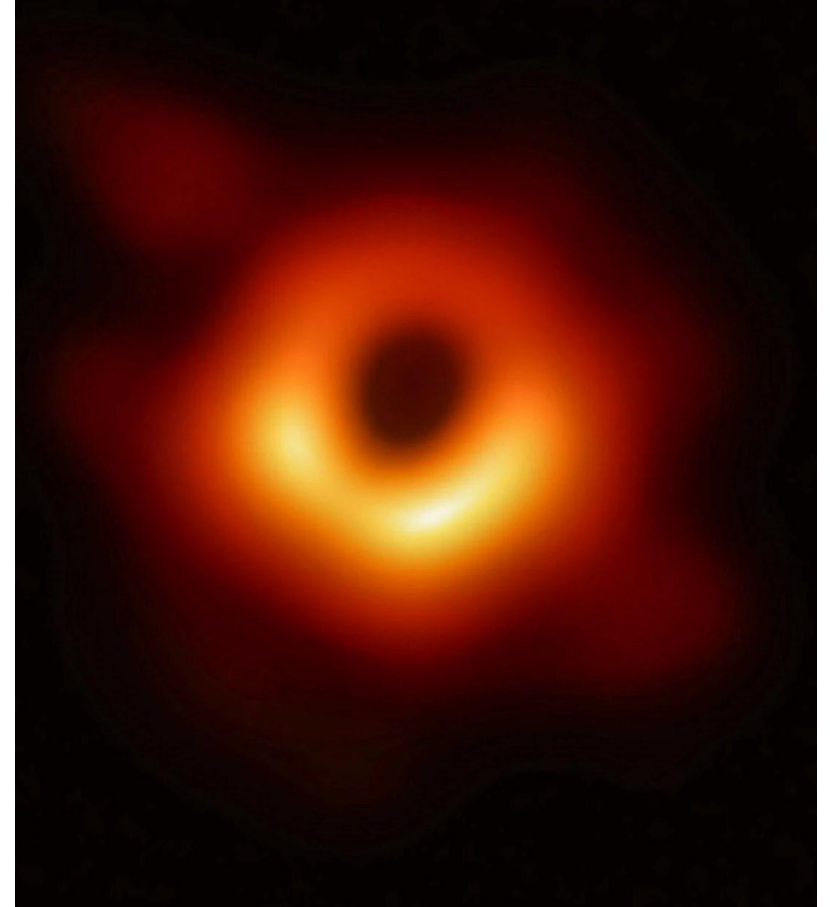
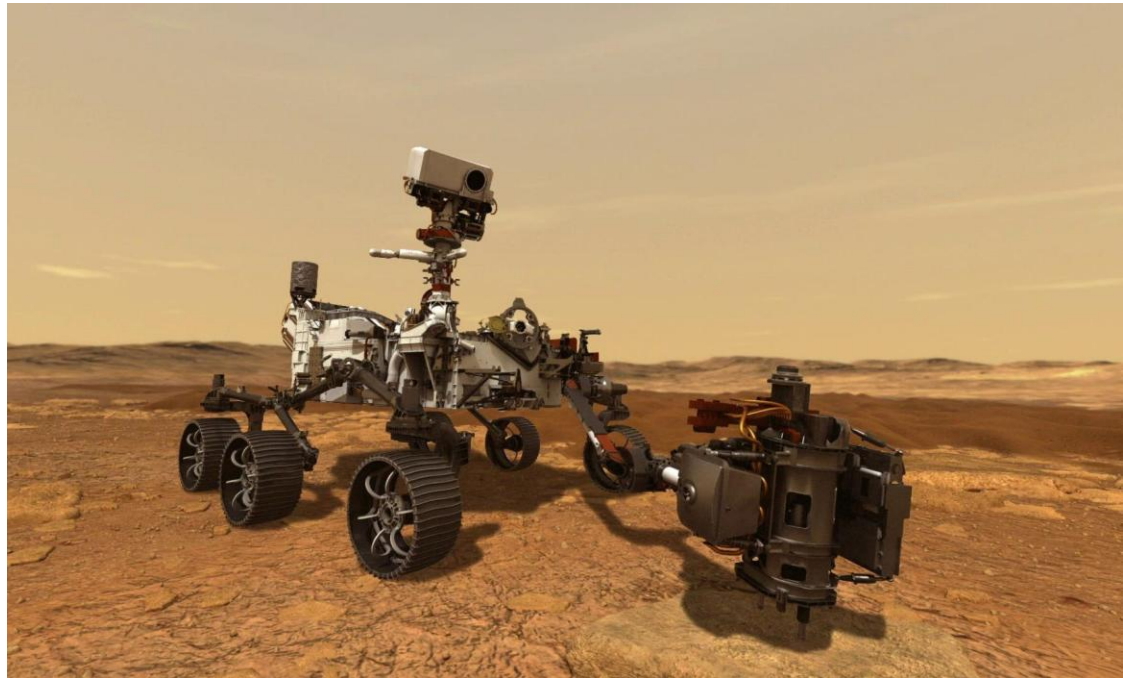
Sweep



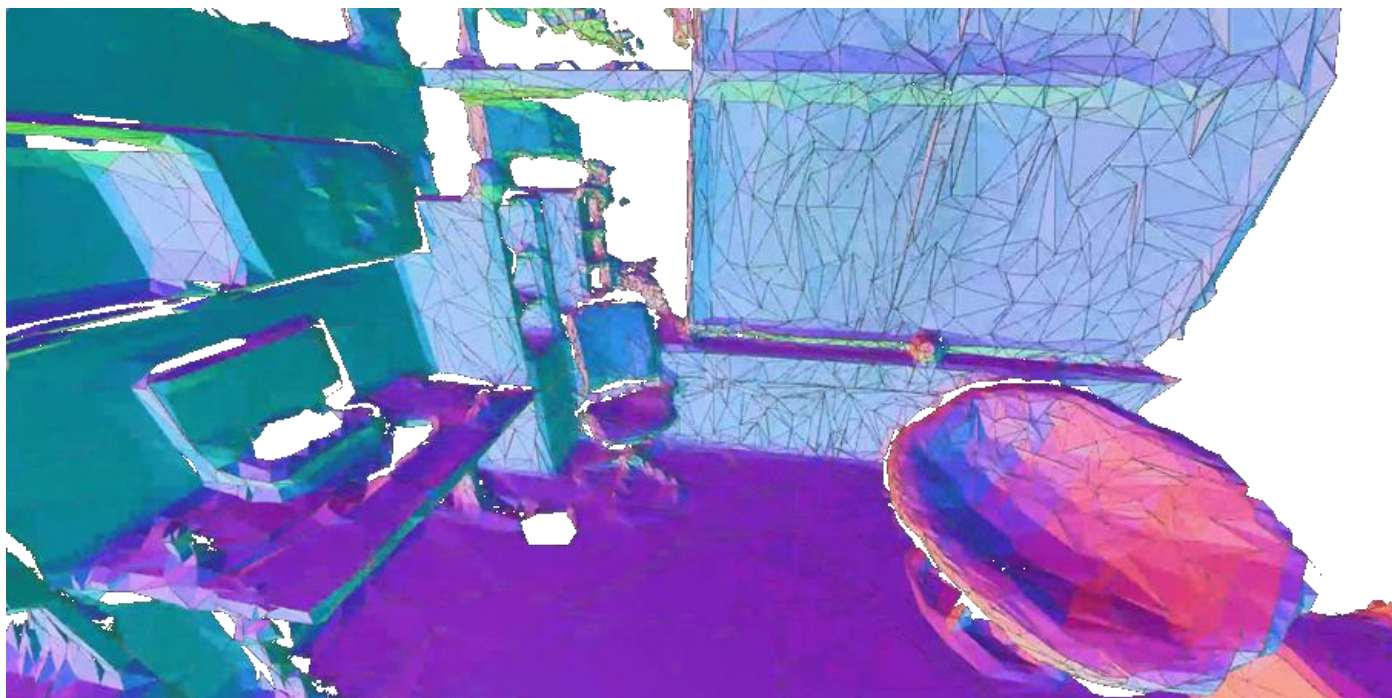
Applications: Security



Application: Space

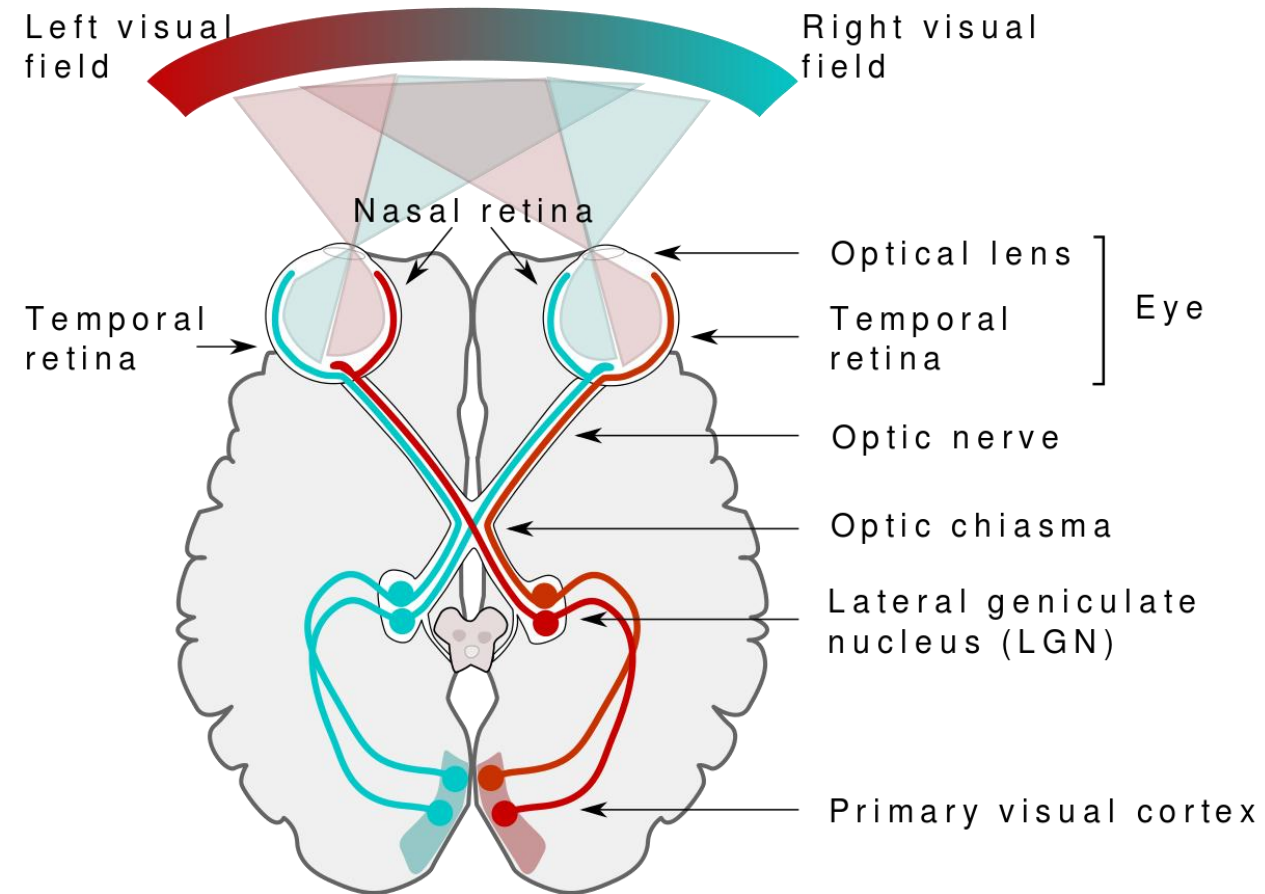
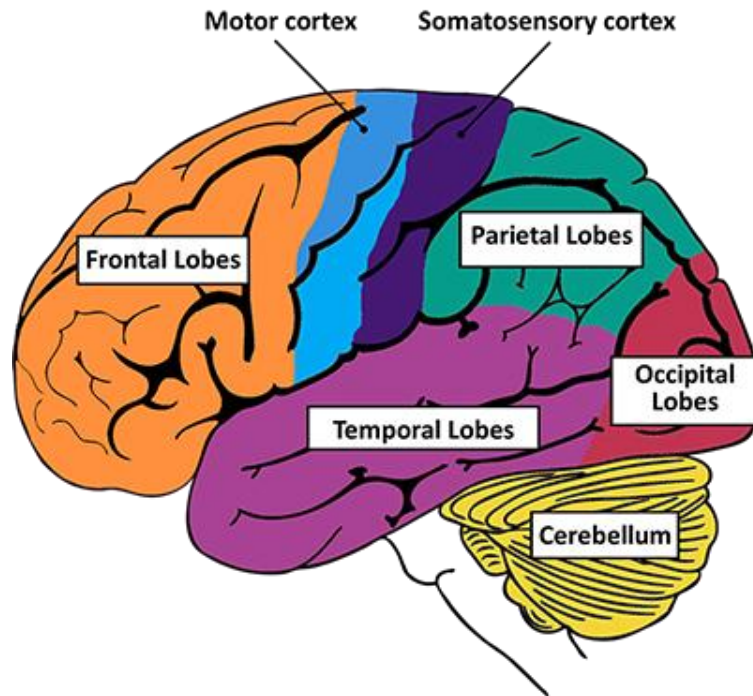


Applications: Augmented Reality

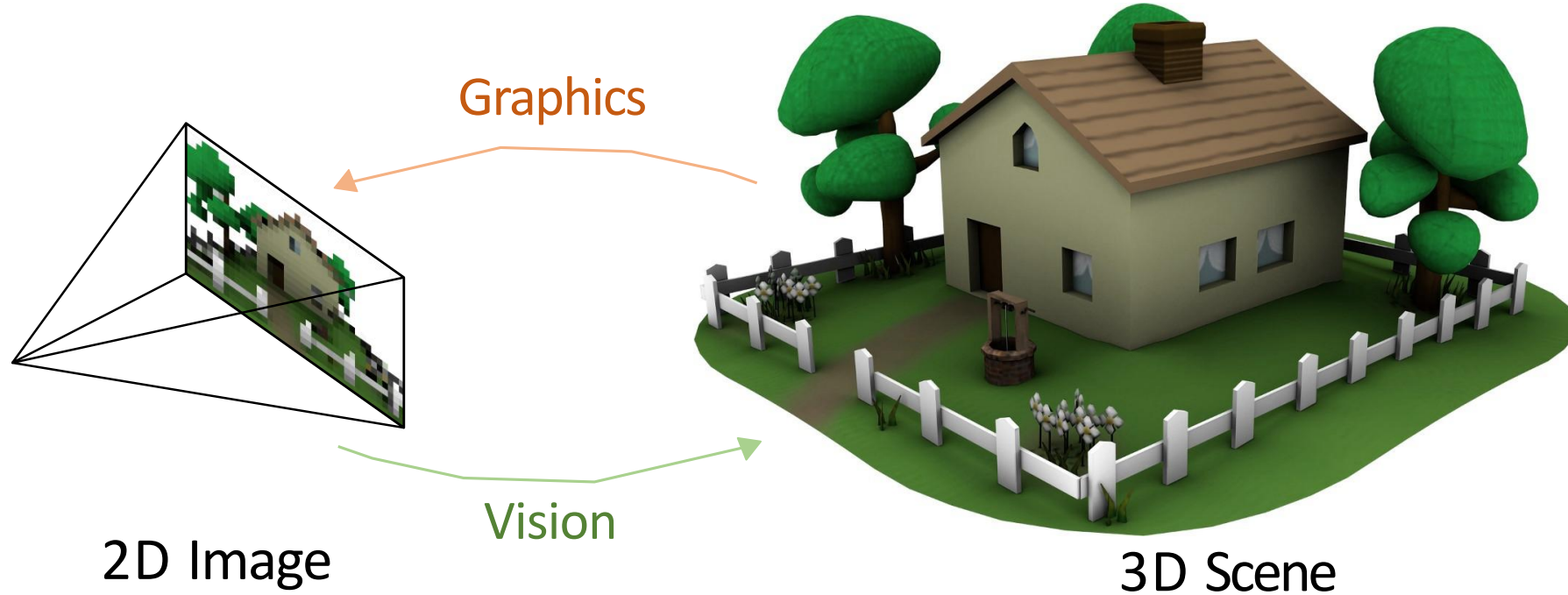


Computer Vision vs. Biological Vision

- Over **50%** of the processing in the **human brain** is dedicated to **visual information**.



Computer Vision vs. Computer Graphics

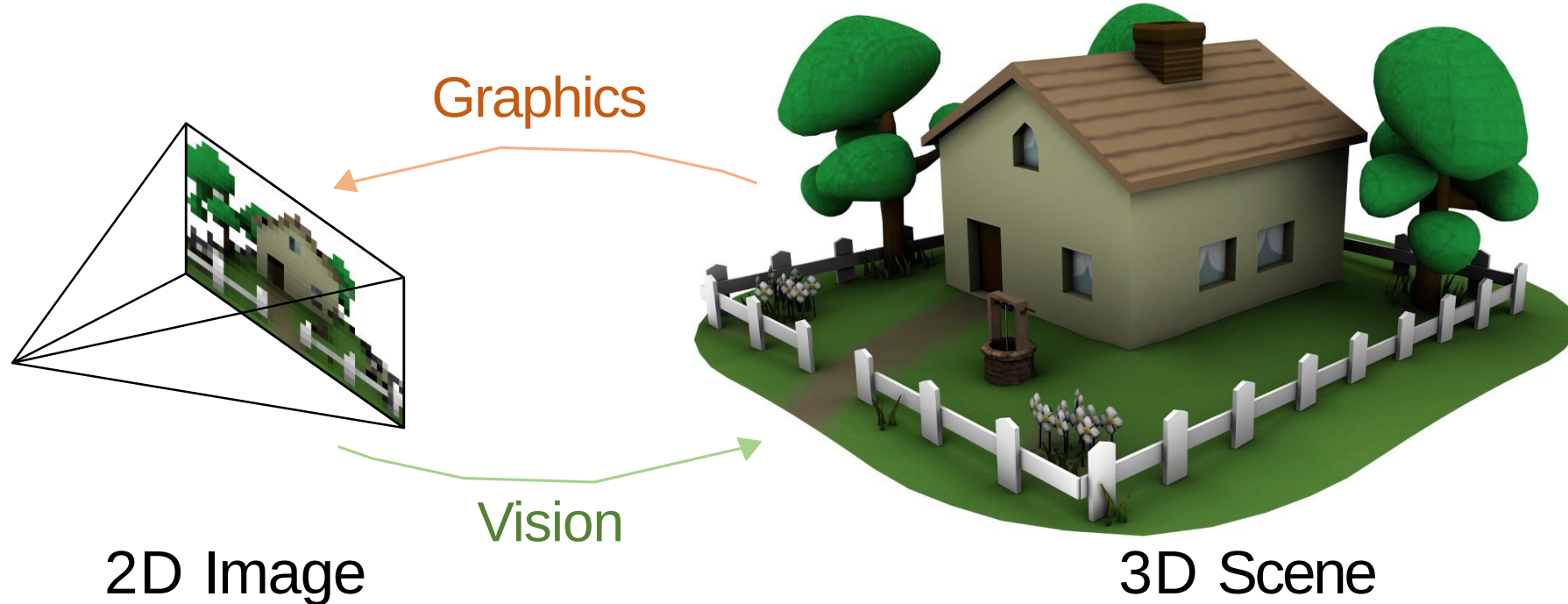


Pixel Matrix

217	191	252	255	239
102	80	200	146	138
159	94	91	121	138
179	106	136	85	41
115	129	83	112	67
94	114	105	111	89

Objects	Material
Shape/Geometry	Motion
Semantics	3D Pose

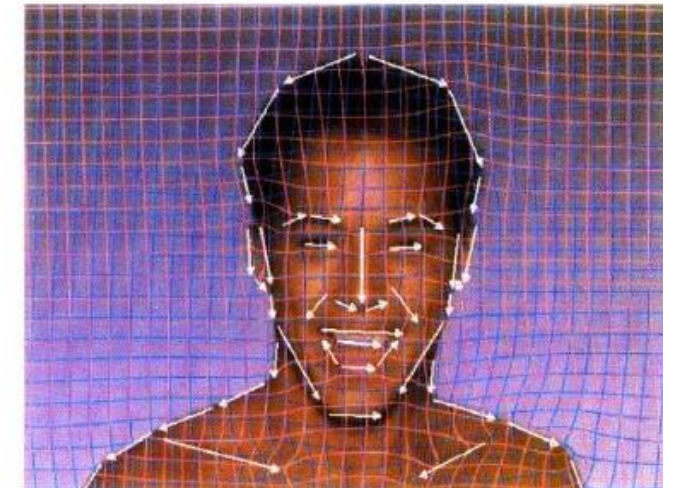
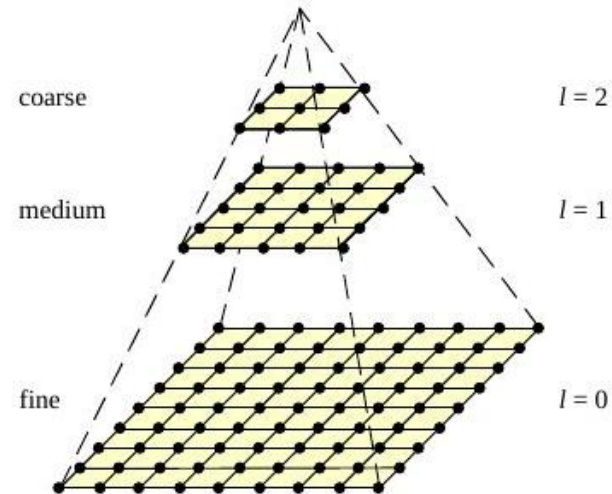
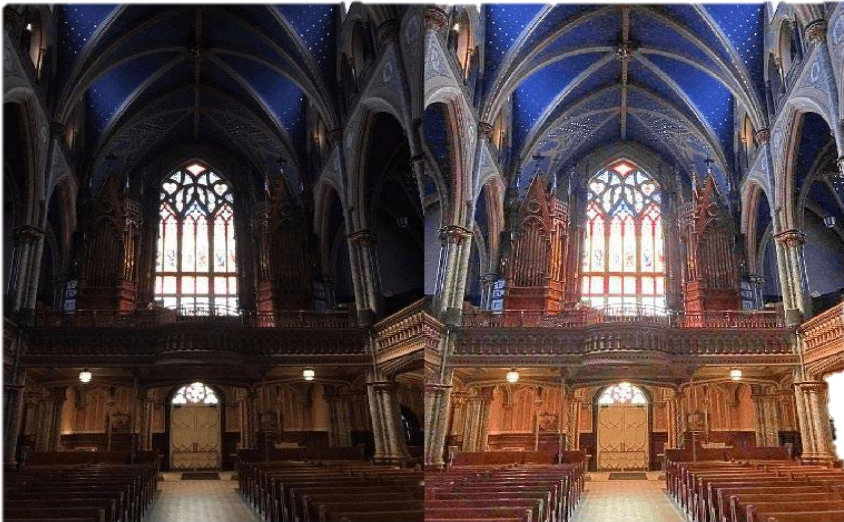
Computer Vision vs. Computer Graphics



Computer Vision is an ill-posed inverse problem

- Many 3D scenes yield the same 2D image
- Additional constraints (knowledge about world) required

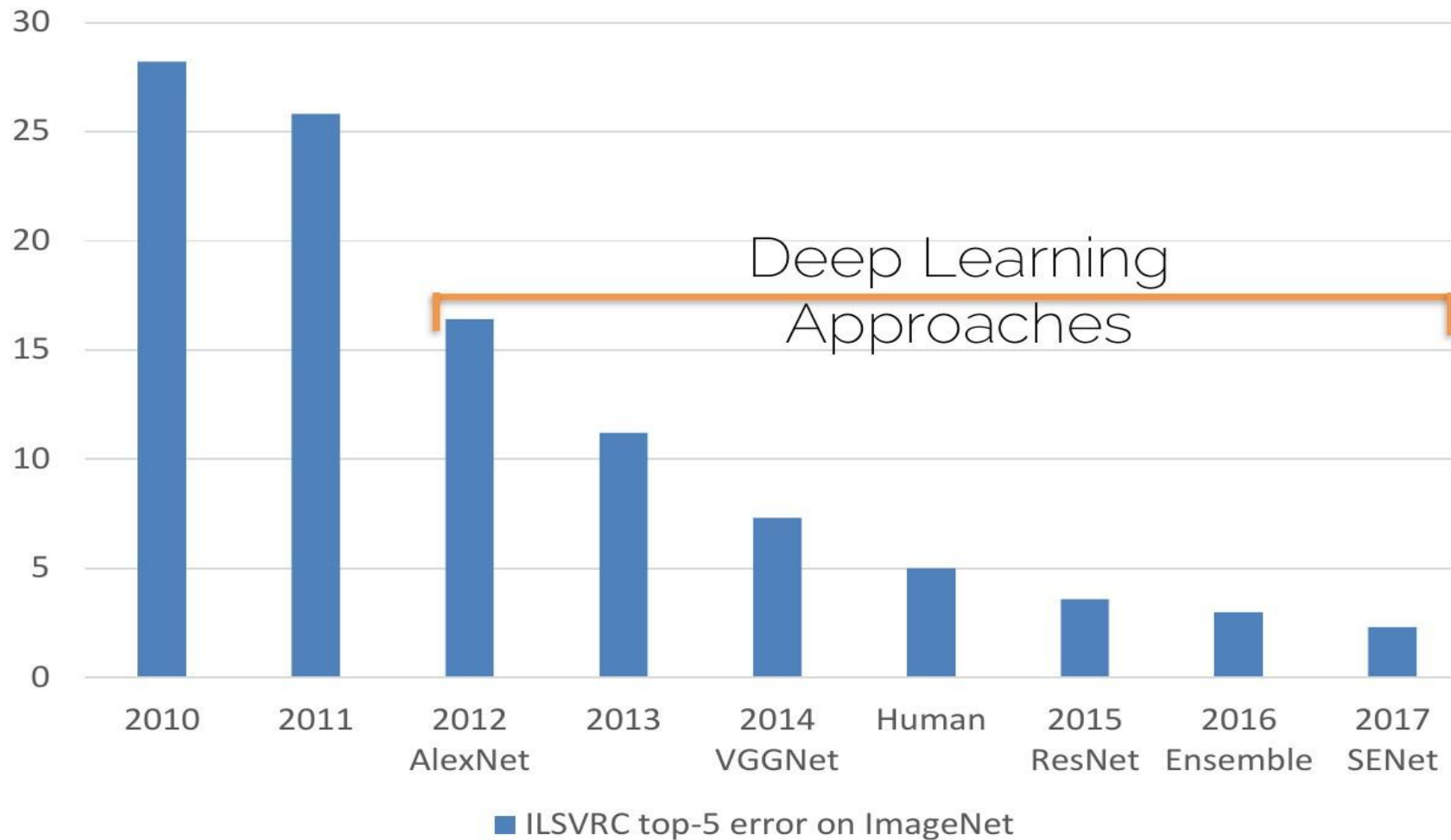
Computer Vision vs. Image Processing



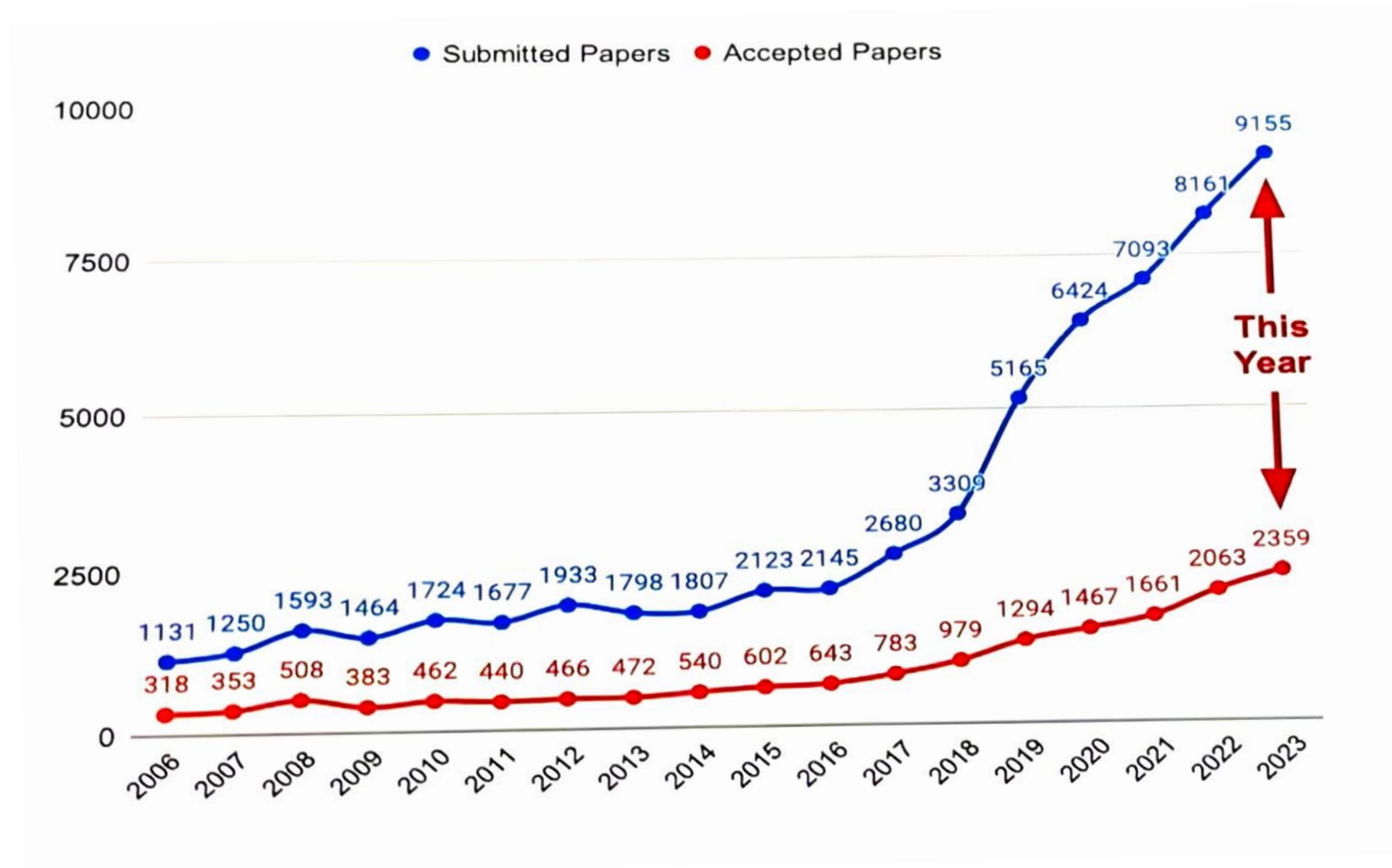
Computer Vision vs. Machine Learning



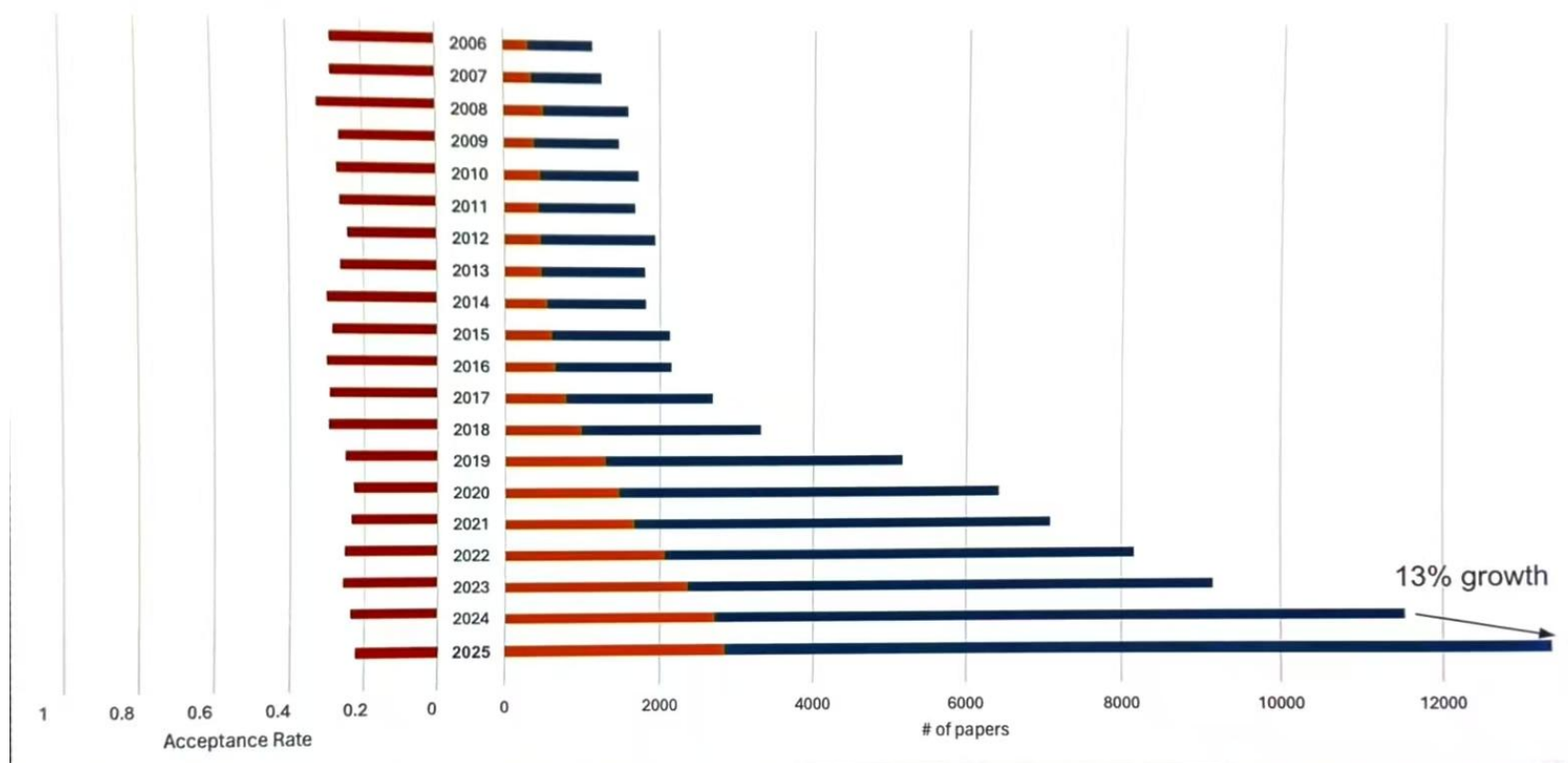
The Deep Learning Revolution



CVPR Submitted and Accepted Papers



New records: 13008 submissions, 2,872 accepted papers!



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What can you learn from this course?

Image Processing

- Image formation
- Image processing
- Image stitching

3D Vision

- Features and matching
- Shape from Shading
- Structure from motion
- Stereo reconstruction

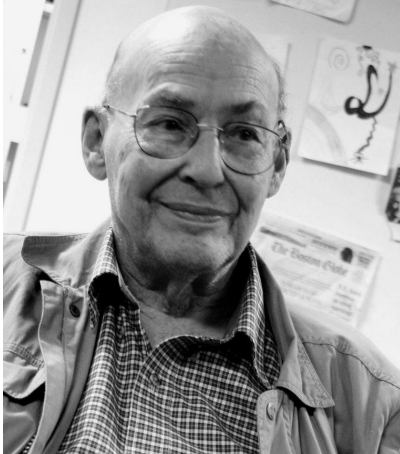
Deep Learning for CV

- Machine learning basics
- CNN and transformers
- Detection, segmentation
- Representation learning



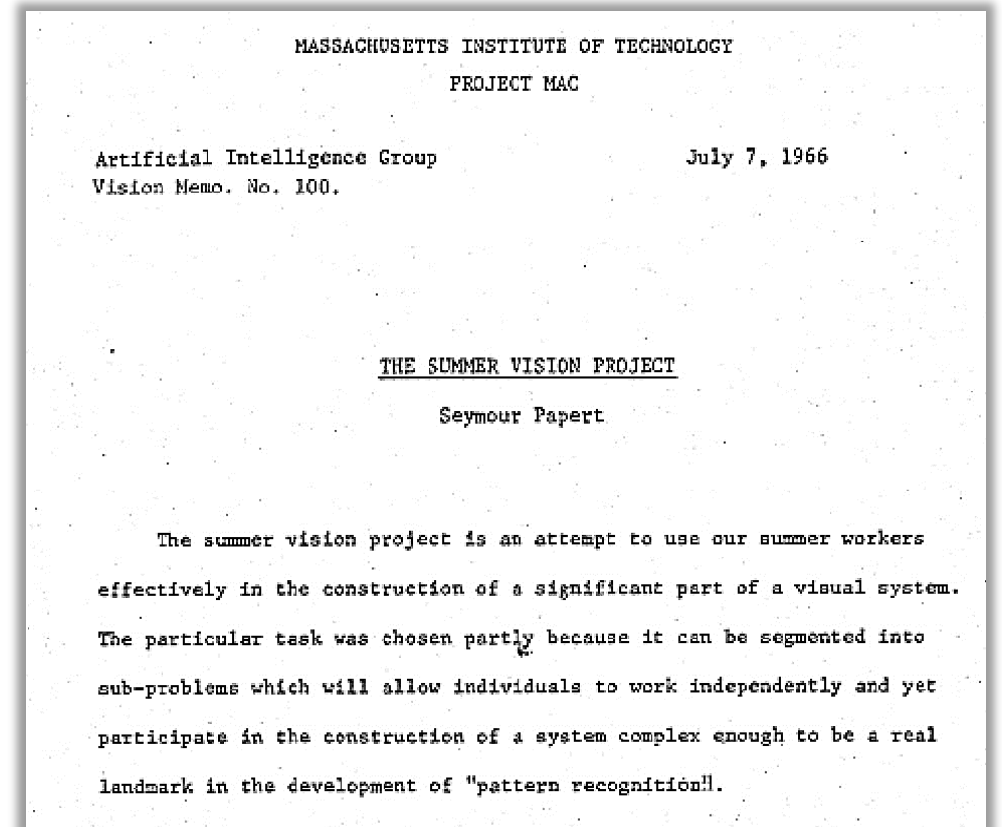
CV is hard

Birth of CV → CV is hard



In 1966, Prof. Marvin Minsky at MIT hired a first-year undergraduate student and assigned him a problem to solve over the summer:

Connect a camera to a computer and get the machine to describe what it sees.



Challenges: Semantic Gap

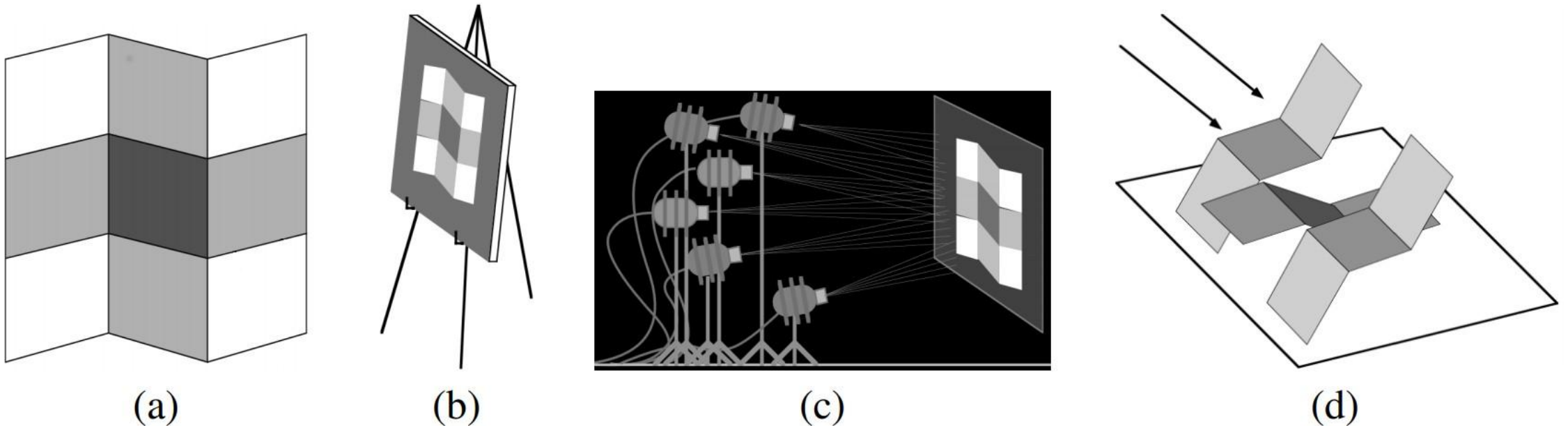


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What the computer sees:

- An image is a grid of numbers in $[0, 255]$
- e.g., $800 \times 600 \times 3$ (3 channels RGB)

Challenges: Images are 2D Projections of the 3D World



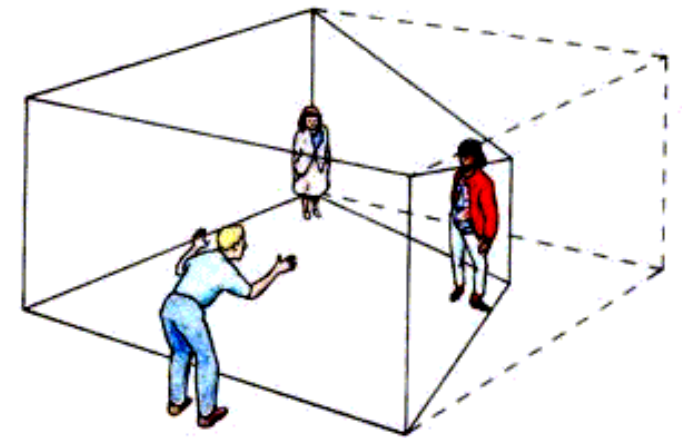
Adelson and Pentland's workshop metaphor:

To explain an image (a) in terms of reflectance, lighting and shape, (b) a painter, (c) a light designer and (d) a sculptor will design three different, but plausible, solutions.

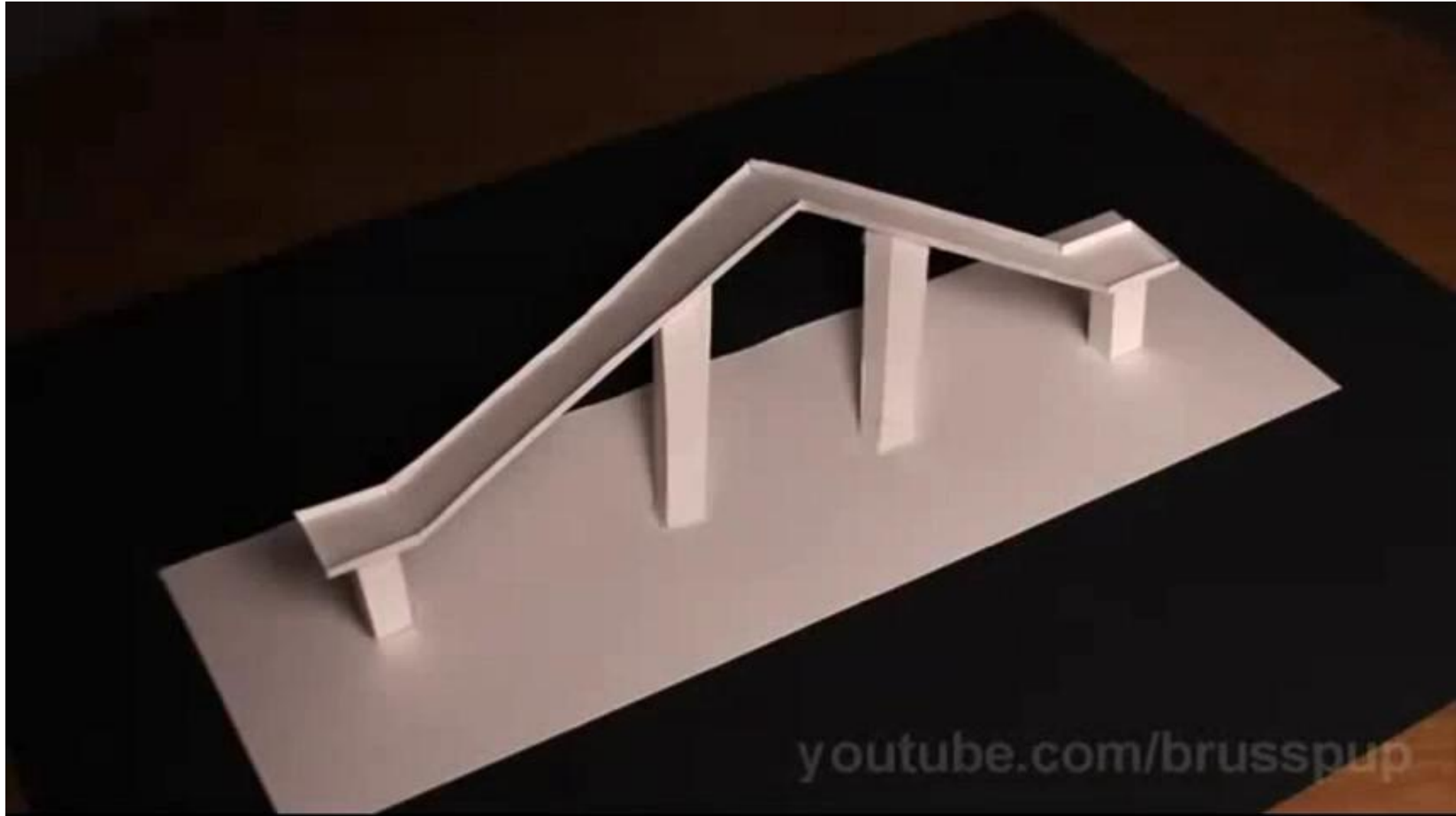
Challenges: Images are 2D Projections of the 3D World



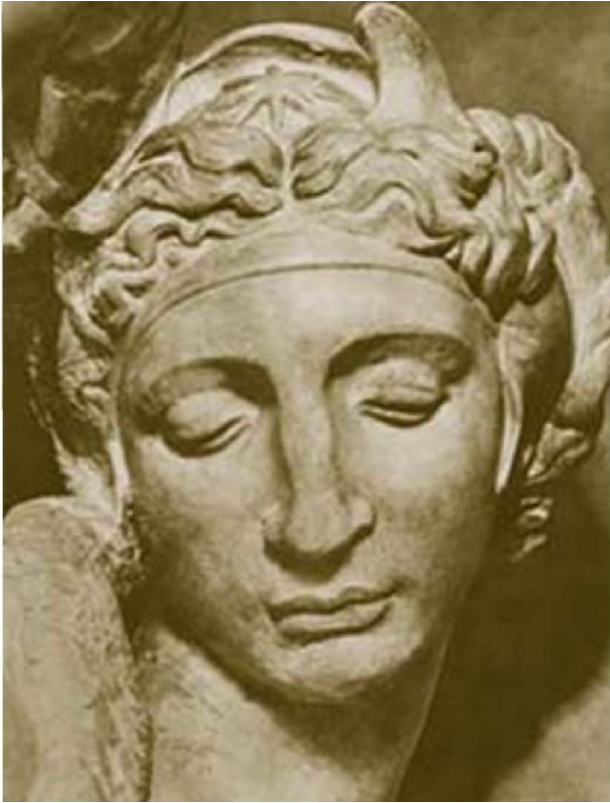
Ames Room Illusion



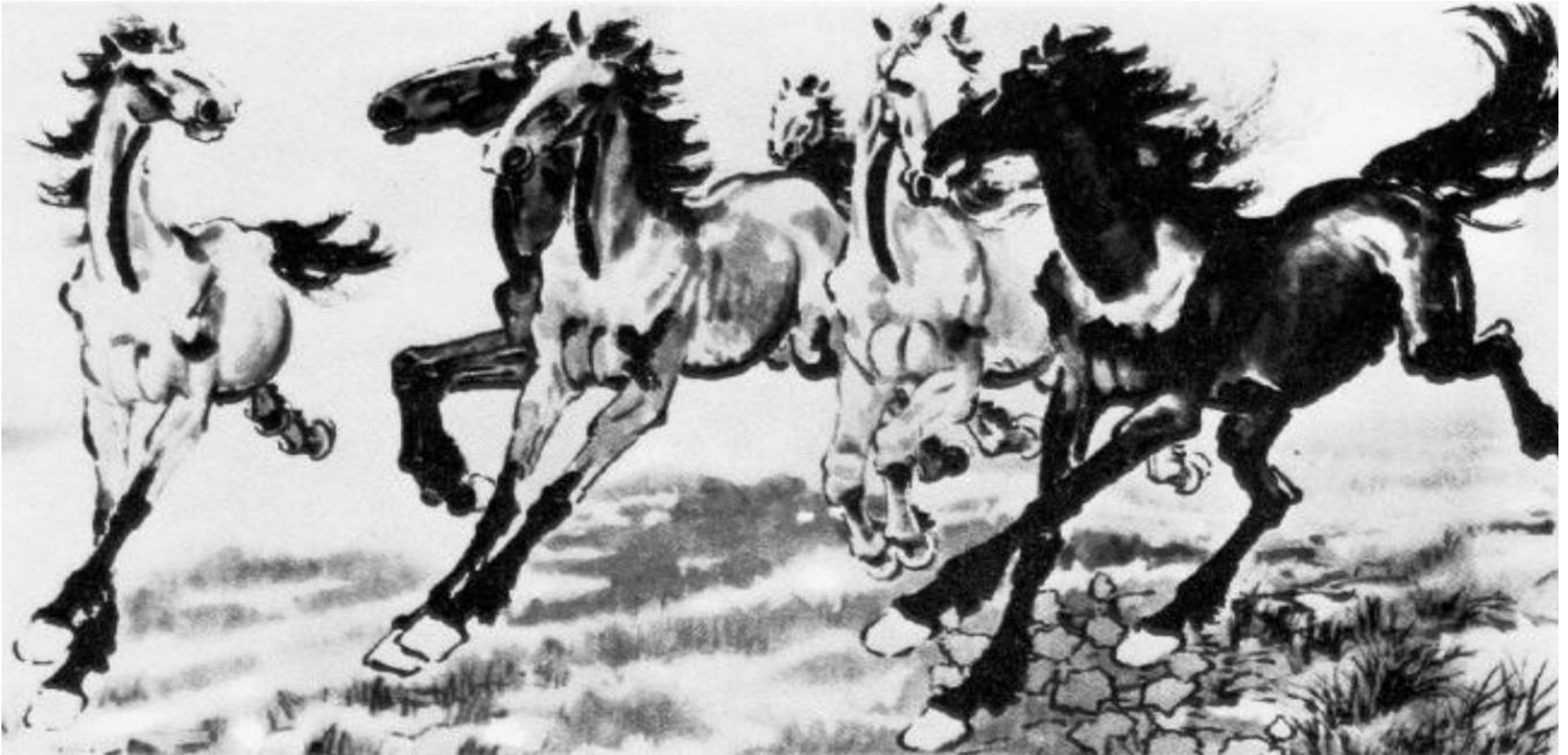
Challenges: Images are 2D Projections of the 3D World



Challenges: Viewpoint Variation



Challenges: Deformation



Xu Beihong (1943)

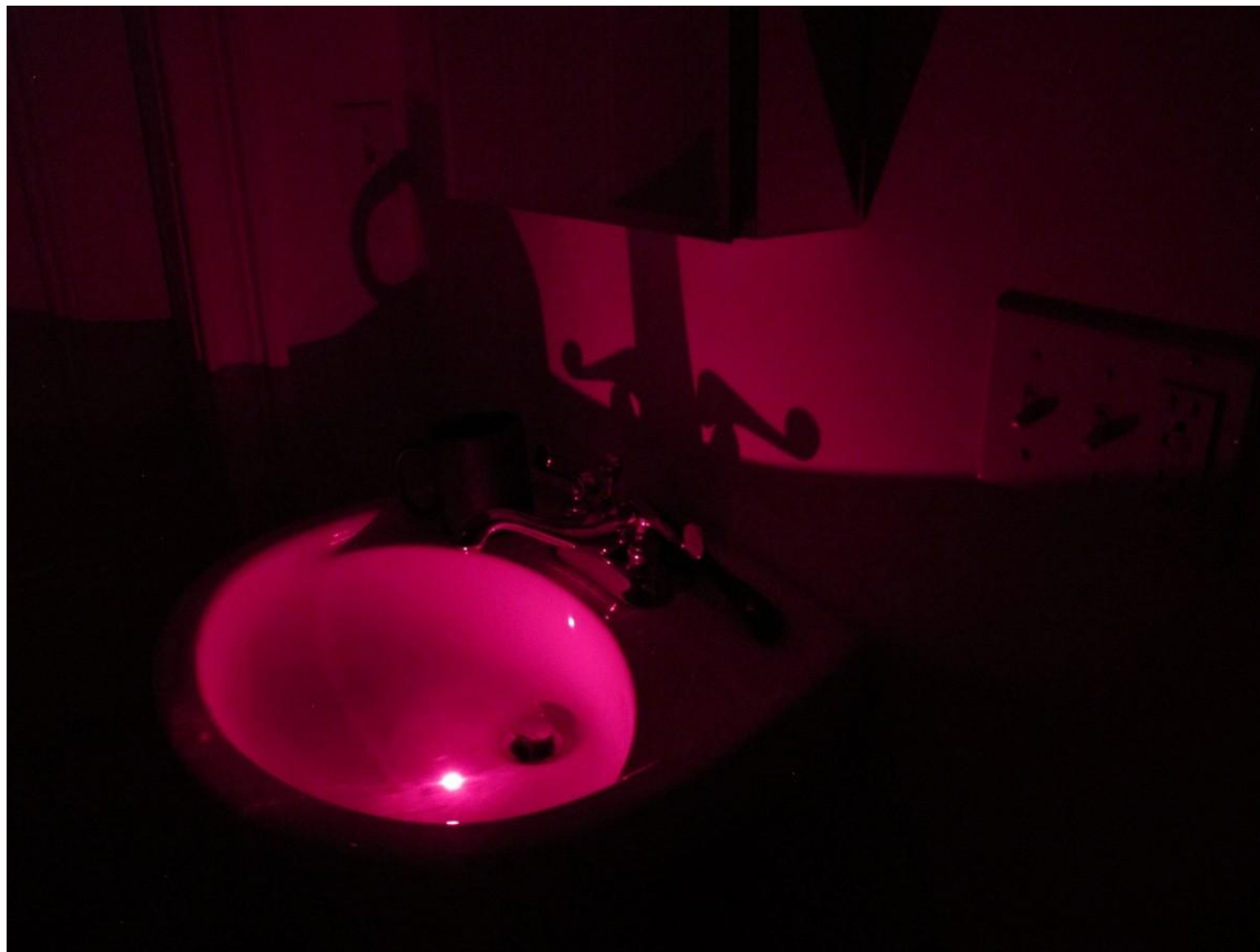
Challenges: Occlusion



Challenges: Illumination



Challenges: Illumination



Challenges: Illumination



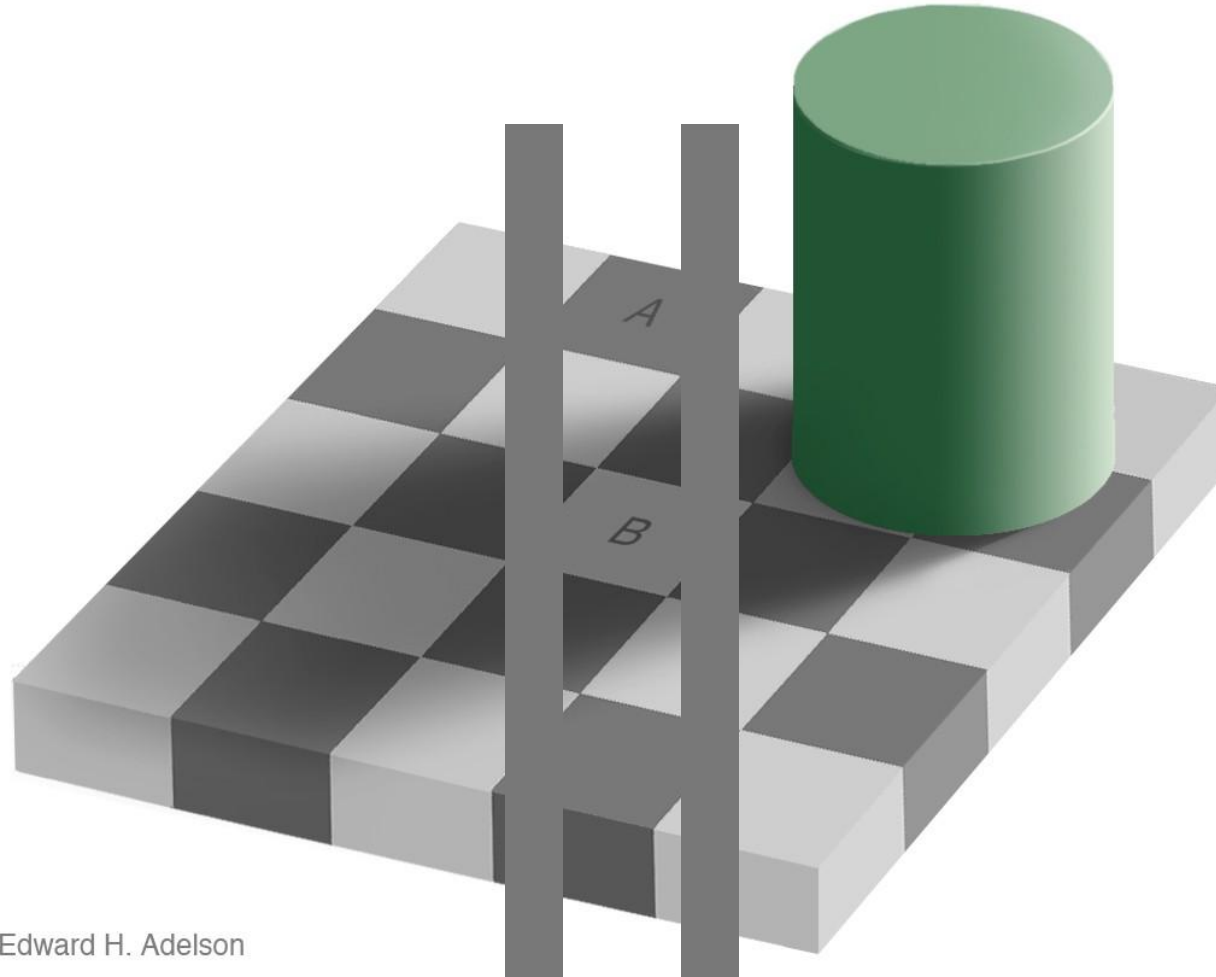
Challenges: Illumination



Challenges: Motion

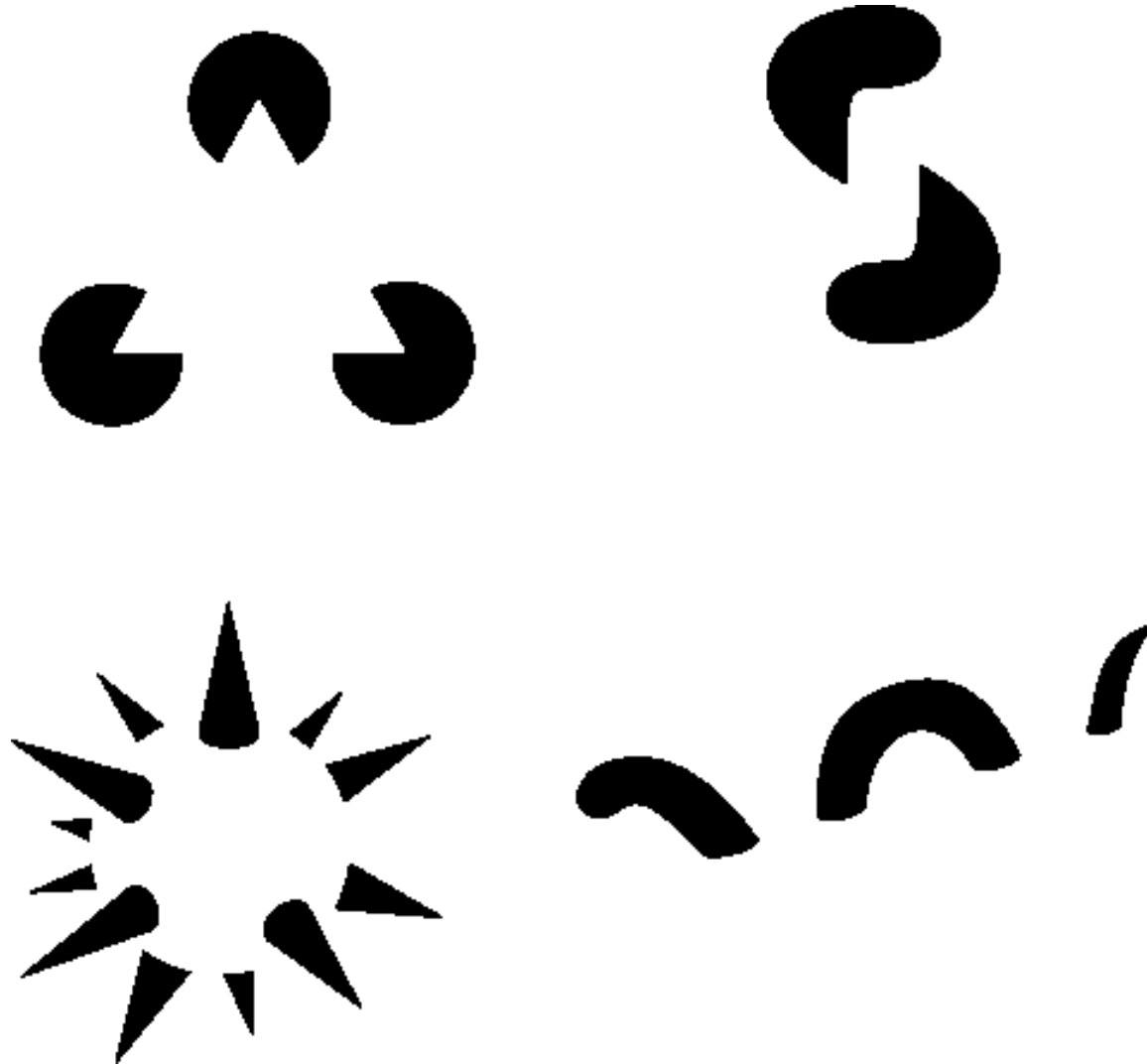


Challenges: Perception vs. Measurement



Edward H. Adelson

Challenges: Perception vs. Measurement



Gaetano Kanizsa (1976)

Challenges: Local Ambiguities



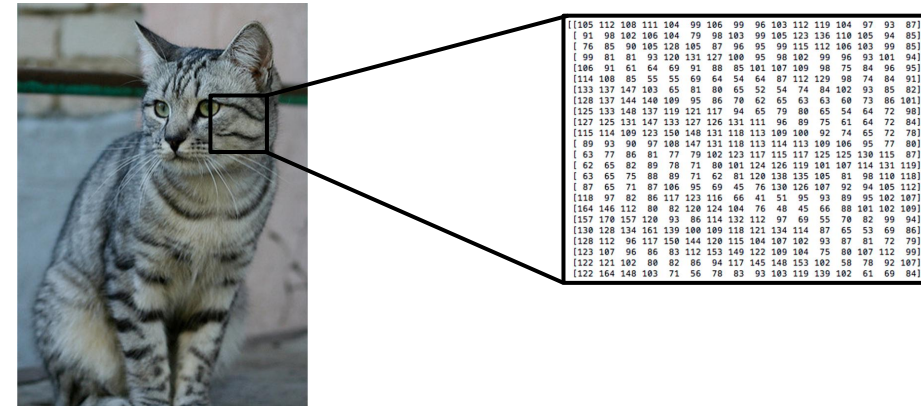
Image Credits: Antonio Torralba

Challenges: Intra Class Variation



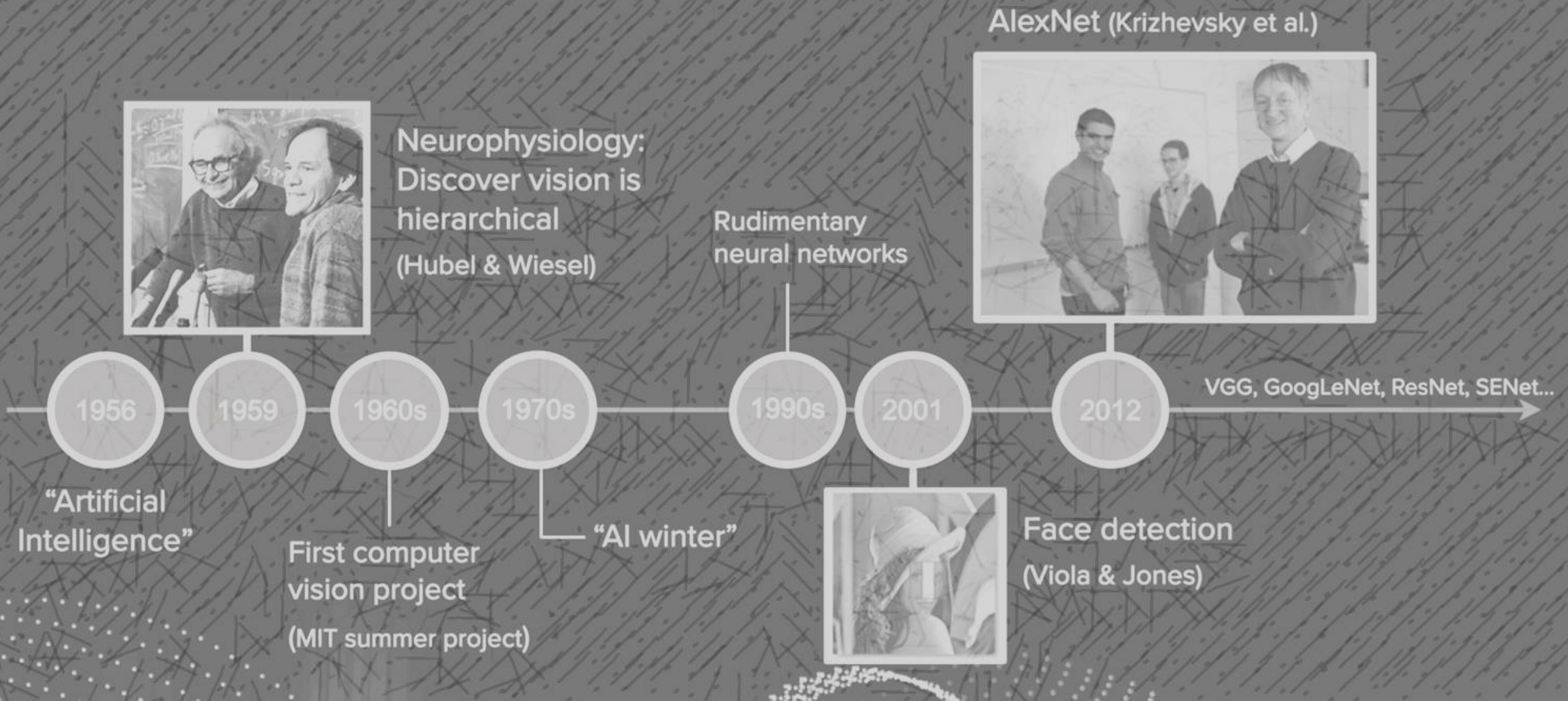
CV is Hard

- Semantic GAP



- Projection, viewpoint, deformation, occlusion, illumination, motion, local ambiguities, variations

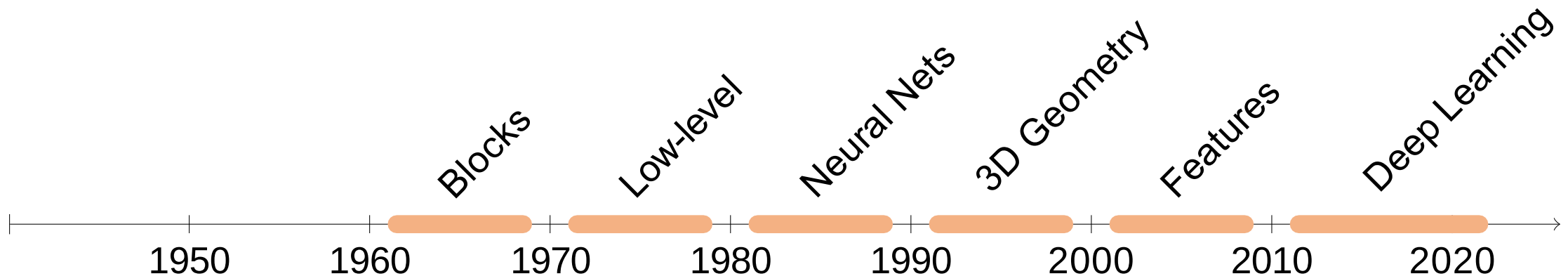




A Brief History of CV

Waves of CV Development

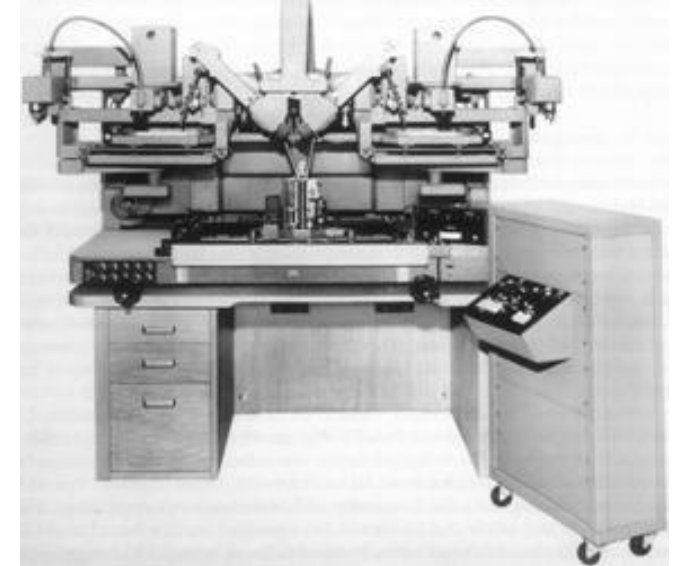
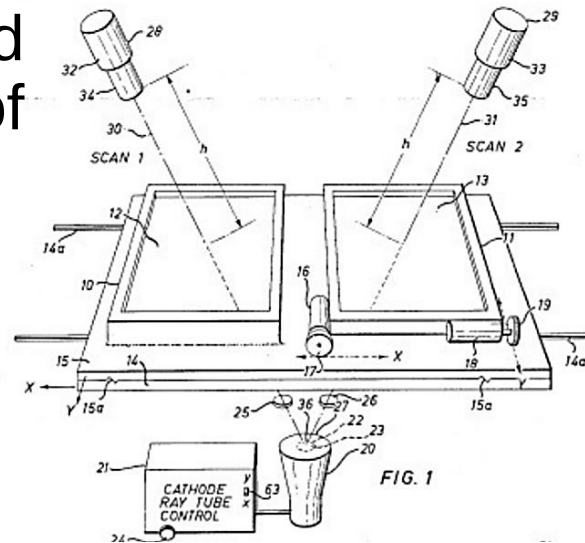
- 1960-1970: **Heuristics vision**: Blocks worlds, edges, and symbolic AI
- 1970-1981: **Low-level vision**: Stereo, flow, shape from shading, Marr's theory
- 1985-1989: **Neural networks**: Backpropagation, early 3D geometry
- 1990-2000: **3D geometry**: Dense stereo, multi-view stereo, LeNet
- 2000-2010: **Feature engineering**: Features, descriptors, structure from motion
- 2010-now : **Deep learning**: Large datasets, deep neural networks



1957: Stereo

Pre-1960 : Early CV

- Gilbert Hobrough demonstrated an analog implementation of stereo image correlation
- 2 transparent images
- 1 illuminator below
- 2 sensors above



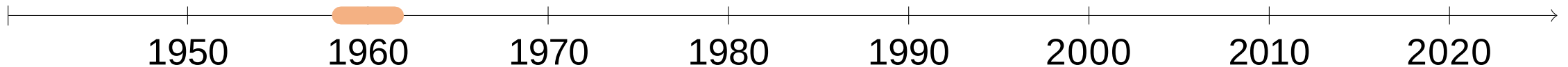
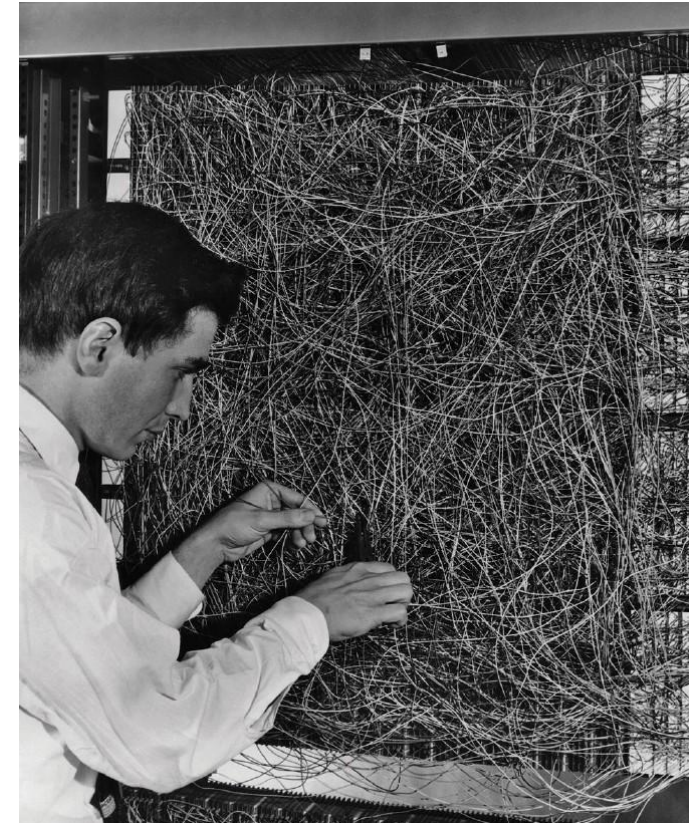
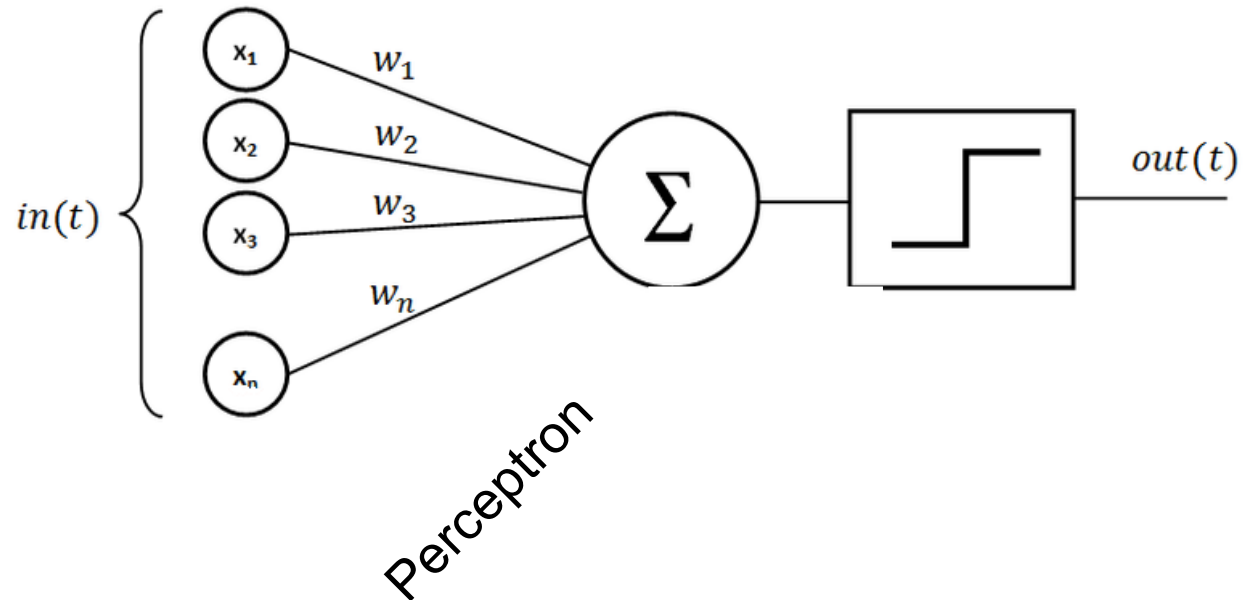
Stereo



1958-1962: Rosenblatt's Perceptron

Pre-1960 : Early CV

- The first algorithm and implementation to train single linear threshold neuron

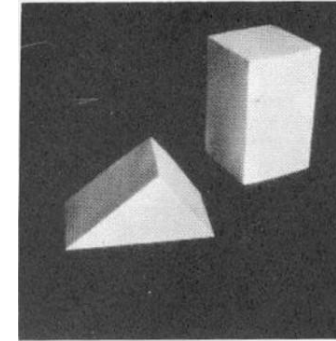


Rosenblatt: The perceptron - a probabilistic model for information storage and organization in the brain, 1958.

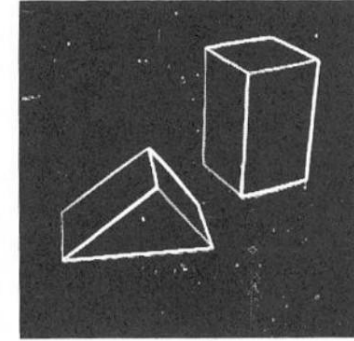
1963: Larry Robert's Blocks World

1960-1970: Heuristics vision

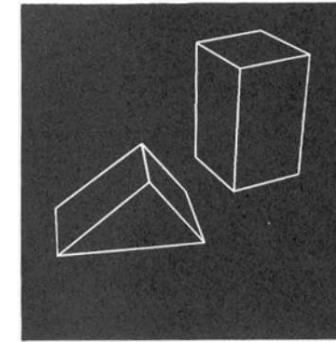
- Scene understanding for robotics
- Extracts edges as primitives
- Infers 3D structure of an object from topological structure of the 2D lines
- Interpret images as projections of 3D scenes, not 2D pattern recognition



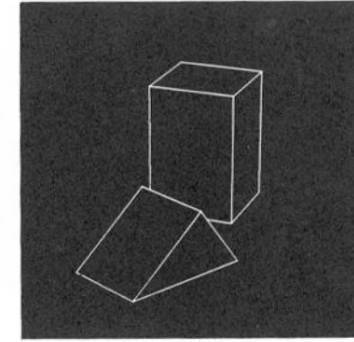
(a) Original picture.



(b) Differentiated picture.

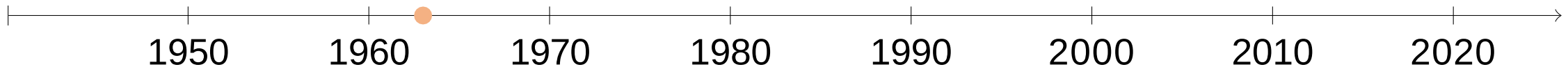


(c) Line drawing.



(d) Rotated view.

Blocks World

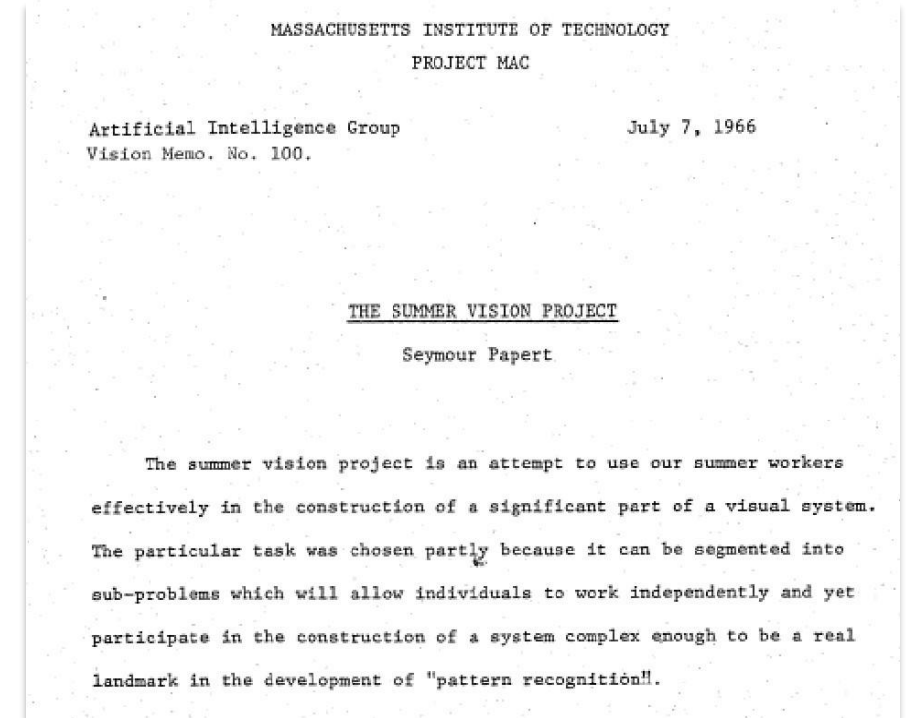


Roberts: Machine Perception of Three-Dimensional Solids. PhD Thesis, 1965.

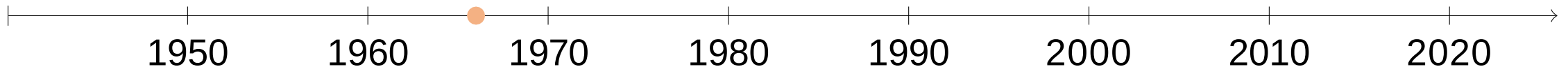
1966: MIT Summer Vision Project

1960-1970: **Heuristics vision**

- Underestimated the challenge of computer vision, committed to “blocks world”



Summer Project

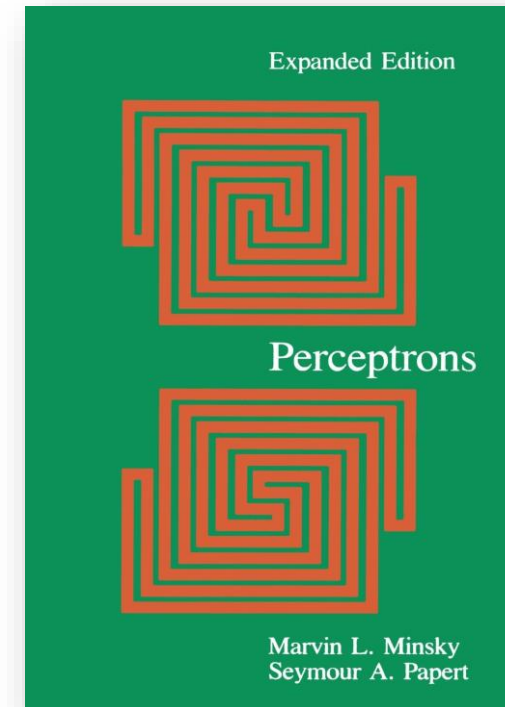


Papert: The Summer Vision Project. MIT AI Memos, 1966.

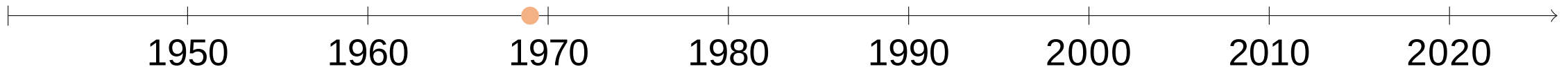
1969: Minsky and Papert Published Perceptrons

1960-1970: **Heuristics vision**

- Showed that single-layer perceptrons cannot solve some very simple problems (XOR problem, counting).
- Symbolic AI research dominated 70s.
- Contributed to the so-called AI winter



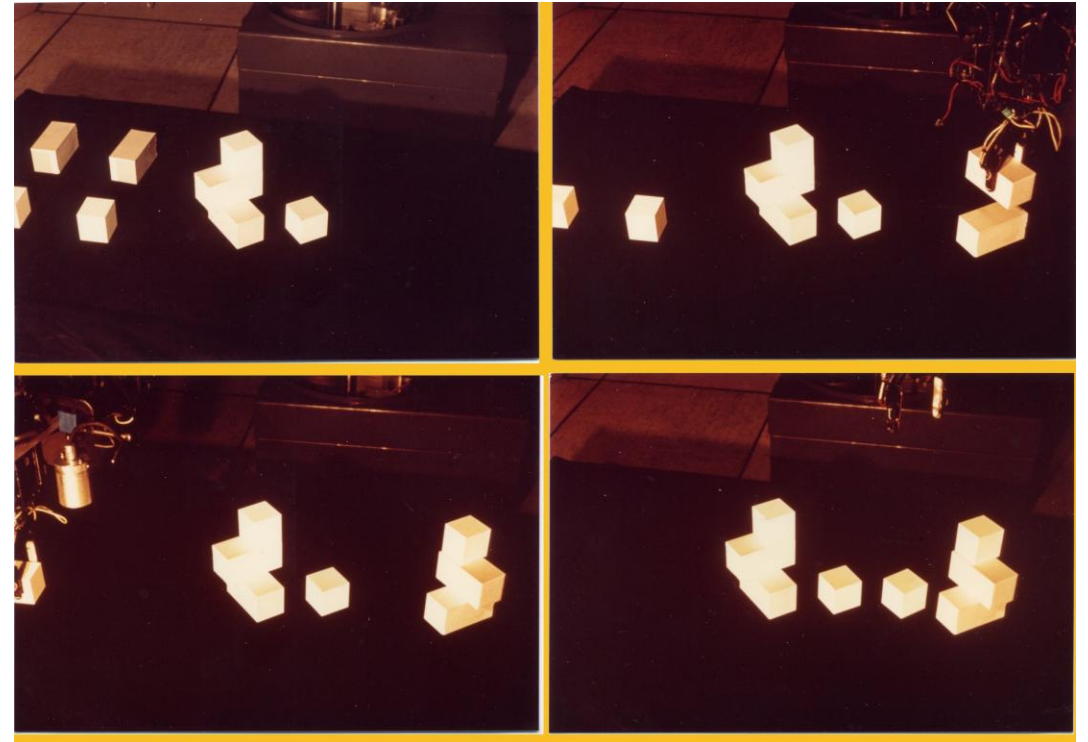
Minsky/Papert



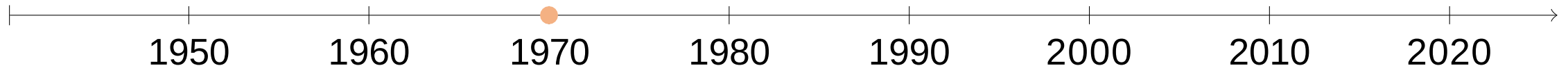
Minsky and Papert: Perceptrons: An introduction to computational geometry. MIT Press, 1969.

1970: MIT AI Lab Copy Demo

- Vision system recovers structure of a scene, robot plans and builds copy from another set of blocks
- Vision, planning and manipulation
- But low-level edge finding is not robust enough, leading to attention on low-level vision



Copy Demo



1970: Shape from Shading

1970-1981: Low-level vision

- Recover 3D from single 2D image
- Assumes Lambertian surface and constant albedo
- Applies smoothness regularization to constrain the ill-posed problem

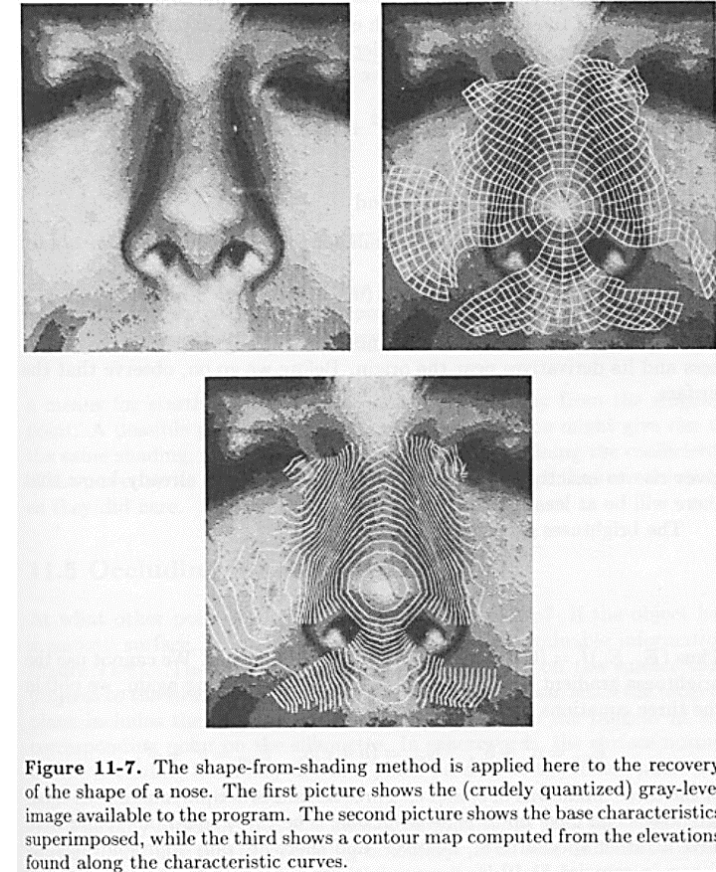
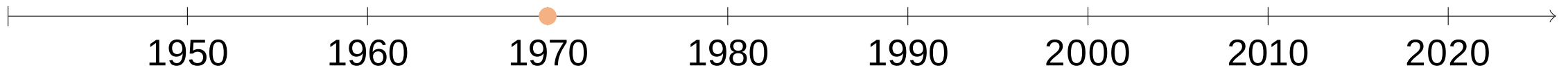


Figure 11-7. The shape-from-shading method is applied here to the recovery of the shape of a nose. The first picture shows the (crudely quantized) gray-level image available to the program. The second picture shows the base characteristics superimposed, while the third shows a contour map computed from the elevations found along the characteristic curves.

SfS

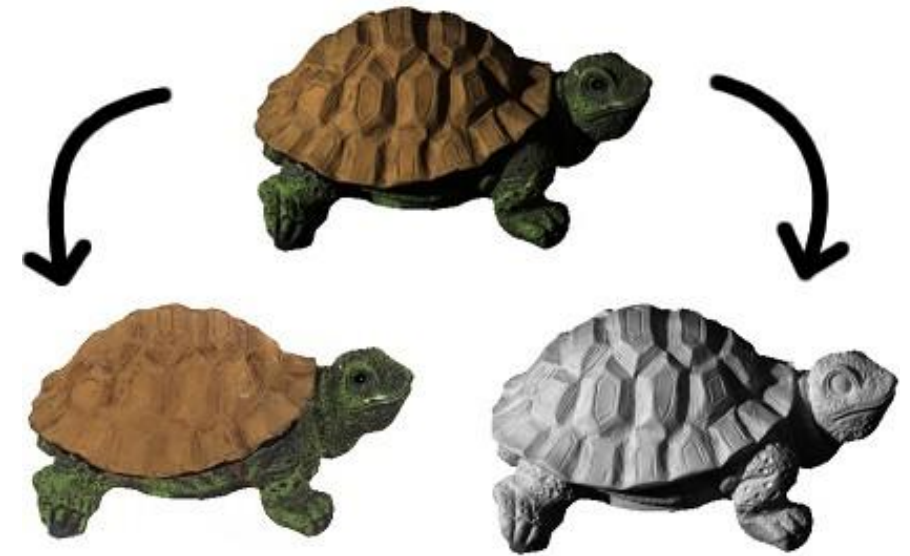


Horn: Shape From Shading: A Method for Obtaining the Shape of a Smooth Opaque Object From One View. MIT TR, 1970.

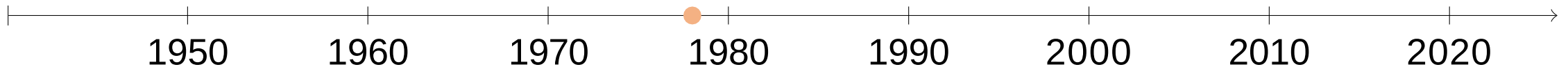
1978: Intrinsic Images

1970-1981: Low-level vision

- Decomposing an image into its different intrinsic 2D layers, such as reflectance, shading, shape and motion components
- Useful for downstream tasks, e.g., object detection independent from shadows and lighting



Intrinsic Images

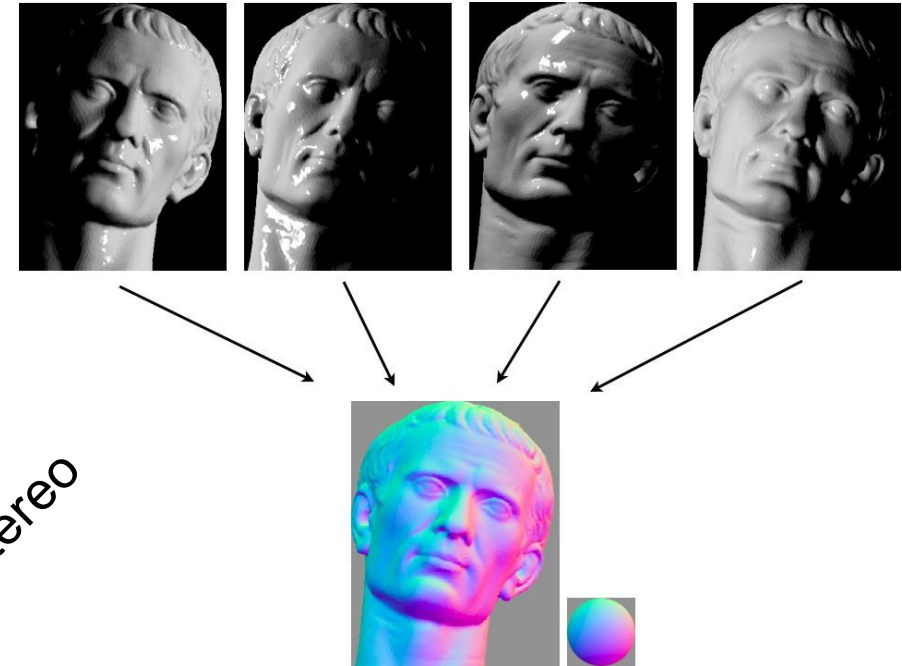


Barrow and Tenenbaum: Recovering intrinsic scene characteristics from images. Computer Vision Systems, 1978.

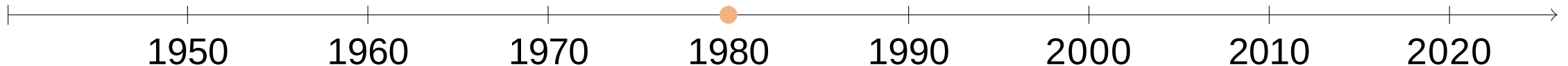
1980: Photometric Stereo

1970-1981: Low-level vision

- Recover 3D from multiple 2D images, taken from the same viewpoint with different lighting conditions
- Requires at least 3 images
- Unprecedented detail and accuracy
- Lambertian assumption has been relaxed subsequently



Photometric Stereo

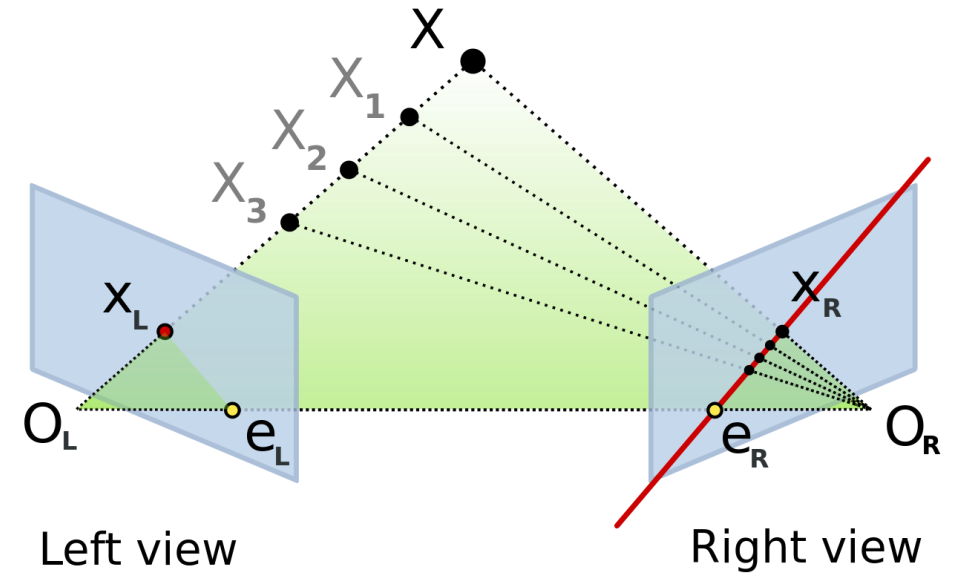


Woodham. Photometric method for determining surface orientation from multiple images. Optical Engineerings, 1980.

1981: Essential Matrix

1970-1981: Low-level vision

- Defines two-view geometry as matrix mapping points to epipolar lines
- Reduces correspondence search to a 1D problem
- Essential matrices can be estimated from a set of 2D correspondences
- Key ideas known for 100 years



Essential Matrix

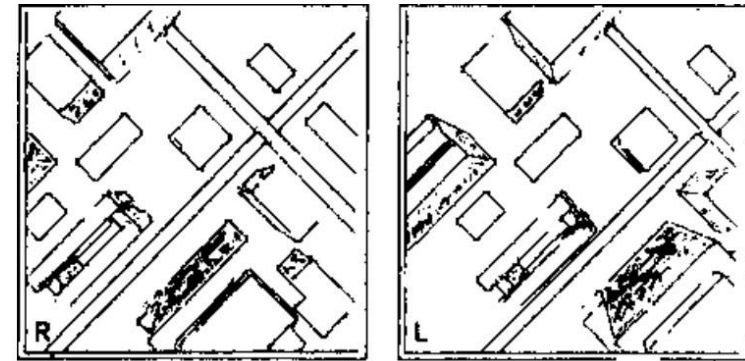


Longuet-Higgins. A computer algorithm for reconstructing a scene from two projections. Nature, 1981.

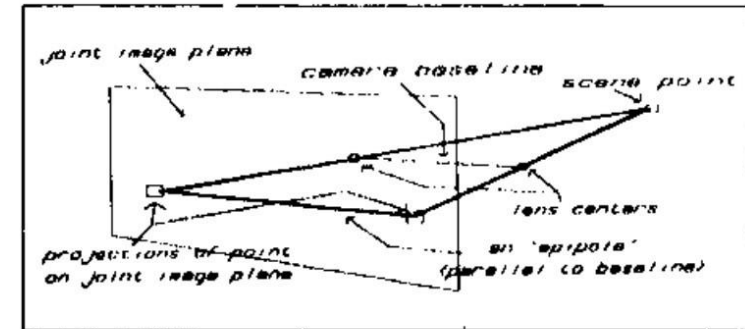
1981: Binocular Scanline Stereo

1970-1981: Low-level vision

- Correlate points along epipolar lines
- Use dynamic programming to introduce constraints along scanlines (image rows)
- Allows for overcoming ambiguities, but streaking artifacts between rows



Edges of the stereo pair
Figure 2



Two Camera Epipolar geometry
Figure 3

Stereo

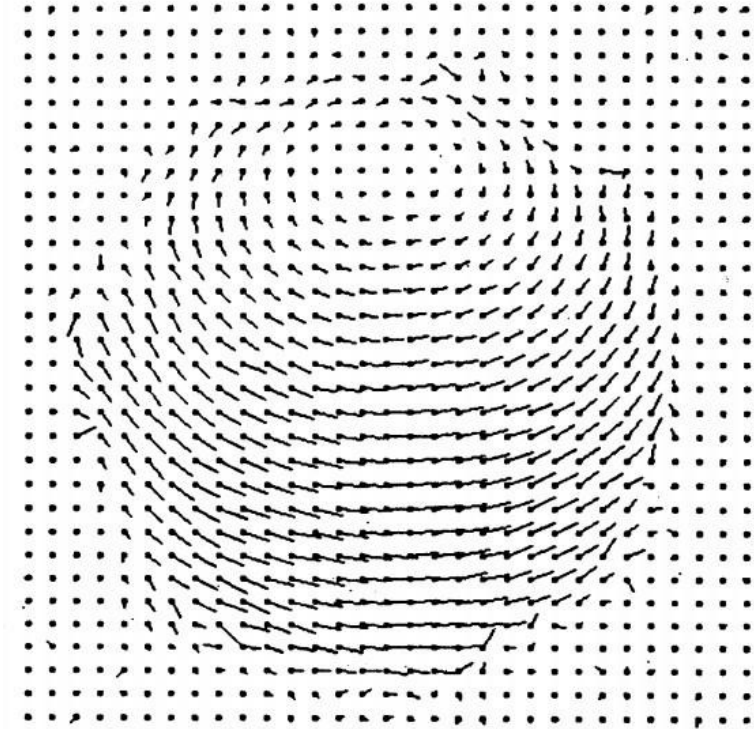


1981: Dense Optical Flow

1970-1981: Low-level vision

- Pattern of apparent motion of objects, surfaces, and edges in a visual scene
- Measured by (densely) tracking pixels between two frames
- Investigated by Gibson to describe the visual stimulus of animals

Optical Flow

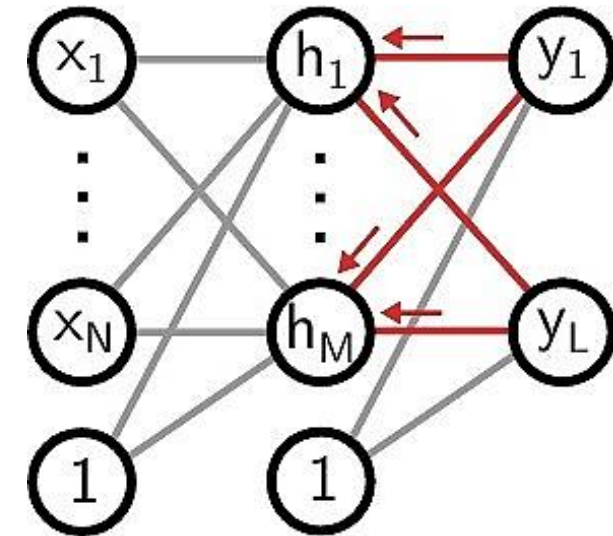


Horn and Schunck: Determining Optical Flow. Artificial Intelligence, 1981.

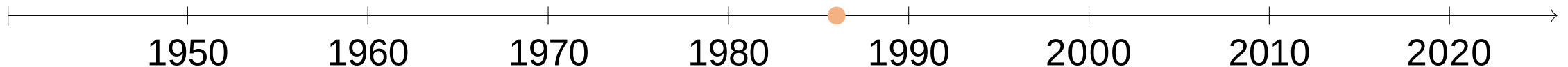
1986: Backpropagation Algorithm

1985-1989: **Neural networks**

- Efficient calculation of gradients in a deep network wrt. network weights
- Enables application of gradient based learning to deep networks
- Known since 1961, but first empirical success in 1986
- Remains main workhorse today



Backpropagation



Rumelhart, **Hinton** and Williams: Learning representations by back-propagating errors. Nature, 1986.

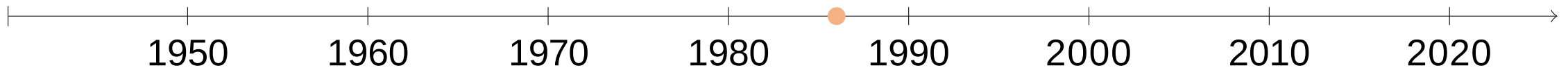
1986: Self-Driving Car VaMoRs

1985-1989: Neural networks

- Developed by Ernst Dickmanns
- Demonstration to Daimler-Benz Research 1986 in Stuttgart
- Longitudinal and lateral guidance with lateral acceleration feedback
- Speed: 0 to 36 km/h



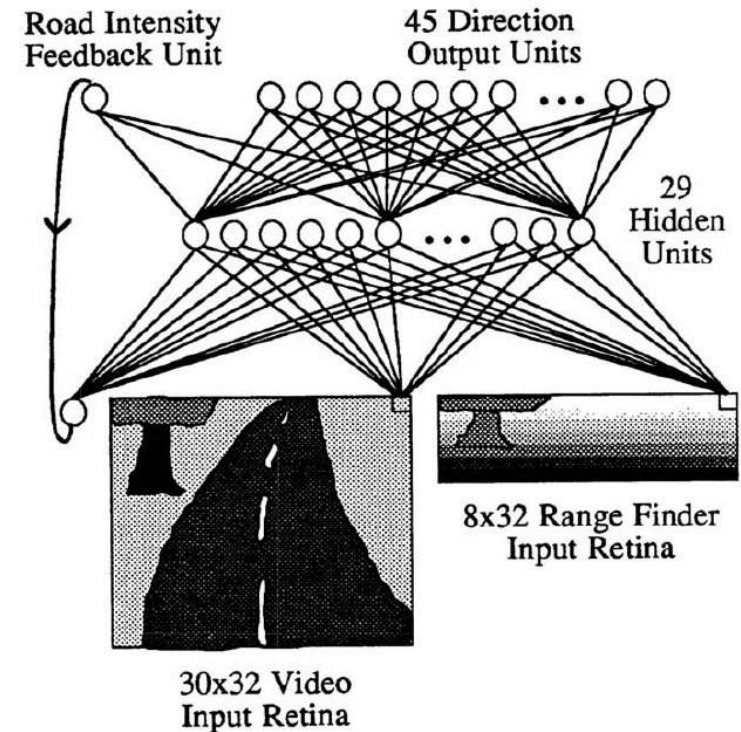
VaMoRs



1988: Self-Driving Car ALVINN

1985-1989: Neural networks

- Forward-looking, vision-based driving
- Fully connected neural network maps road images to vehicle turn radius
- Trained on simulated road images
- Tested on unlined paths, lined city streets and interstate highways
- 90 consecutive miles at up to 70 mph



ALVINN

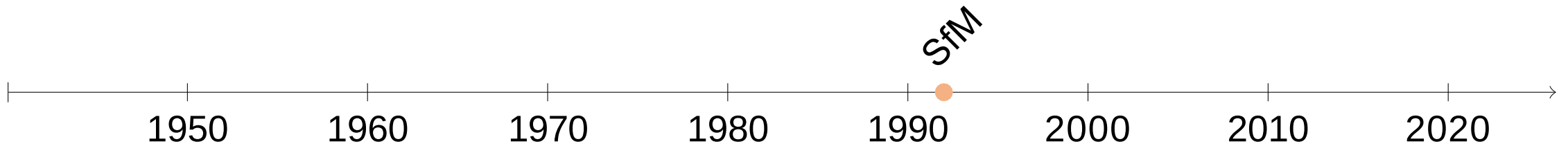
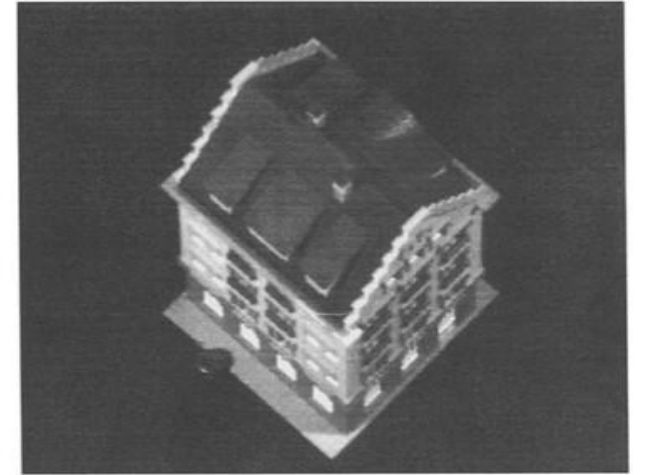


Pomerleau: ALVINN: An Autonomous Land Vehicle in a Neural Network. NIPS, 1988.

1992: Structure from Motion

1990-2000: 3D geometry

- Estimating 3D structures from 2D image sequences of static scenes
- Requires only a single camera
- Tomasi-Kanade factorization provides closed-form (SVD-based) solution for orthographic case

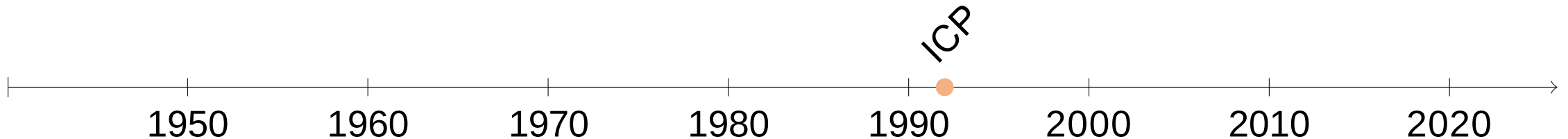
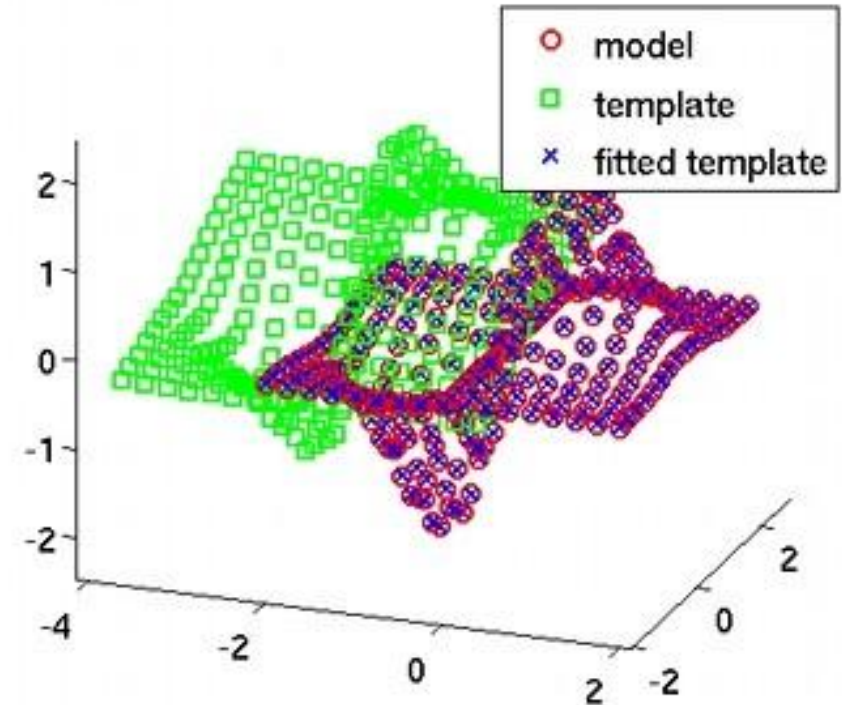


Tomasi and Kanade: Shape and motion from image streams under orthography: a factorization method. IJCV, 1992.

1992: Iterative Closest Points

1990-2000: 3D geometry

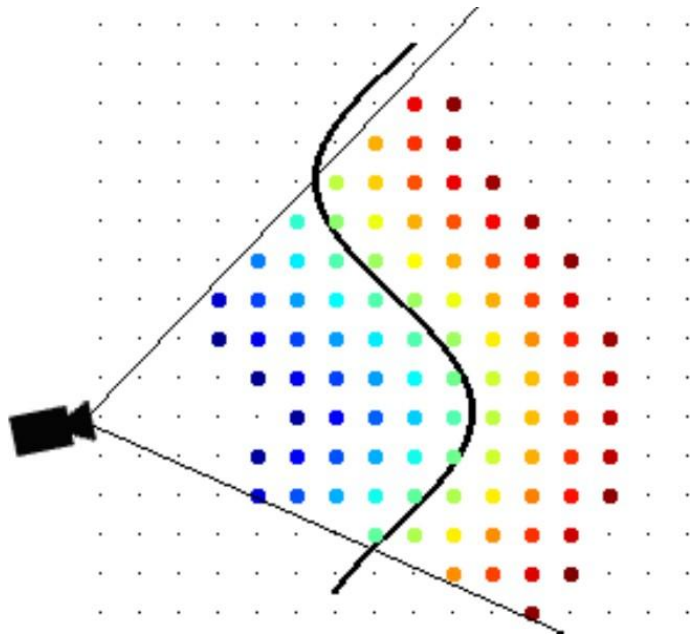
- Registering two point clouds by iteratively optimizing a (rigid or non-rigid) transformation
- Used to aggregate partial 2D or 3D surfaces from different scans, to estimate relative camera poses from point clouds or to localize wrt. a map



Besl and McKay: A Method for Registration of 3-D Shapes. PAMI, 1992.

1996: Volumetric Fusion

1990-2000: 3D geometry



- Aggregation of multiple implicitly represented surfaces by averaging signed distance values
- Mesh-extraction as post-processing

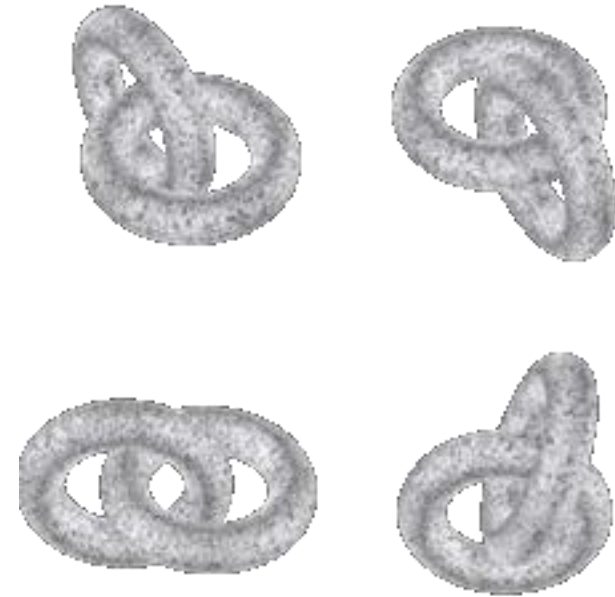


Curless and Levoy: A Volumetric Method for Building Complex Models from Range Images. SIGGRAPH, 1996.

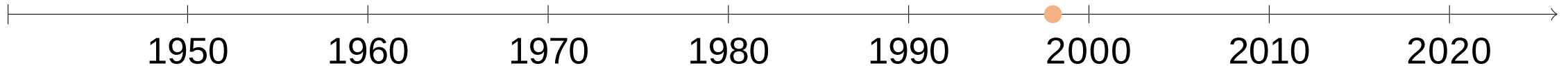
1998: Multi-View Stereo

1990-2000: 3D geometry

- 3D reconstruction from multiple input images using level-set methods
- Flexible topology
- Provable convergence
- Other approaches: Voxel coloring, space carving



MVS

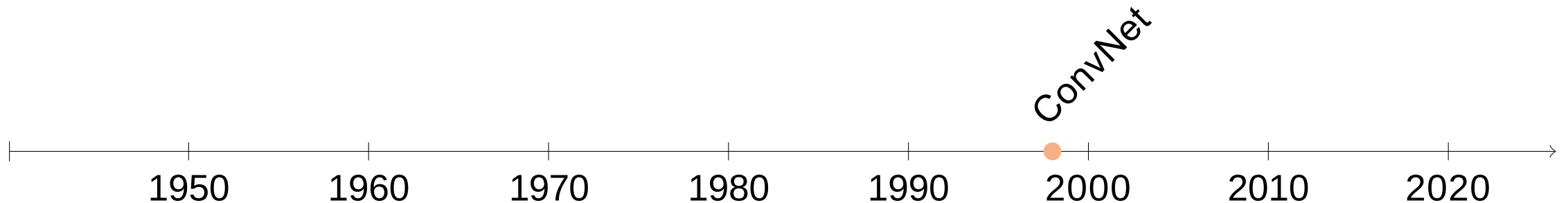
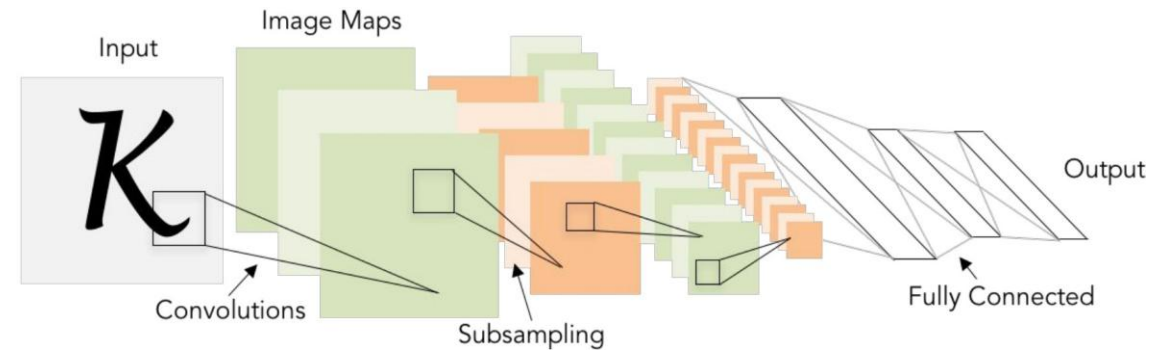


Faugeras and Keriven: Complete Dense Stereovision Using Level Set Methods. ECCV, 1998.

1998: Convolutional Neural Networks

1990-2000: 3D geometry. *Persistence in exploring neural networks.*

- The network is trained end-to-end using backpropagation
- Implements spatial invariance via **convolutions** and max-pooling
- Weight sharing reduces parameters
- Tanh activations
- Only good results on MNIST in 1998

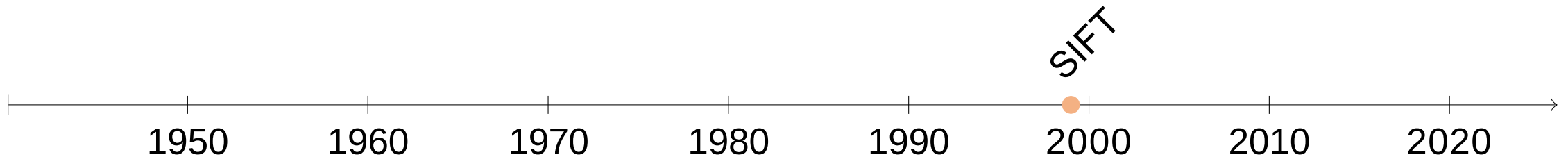
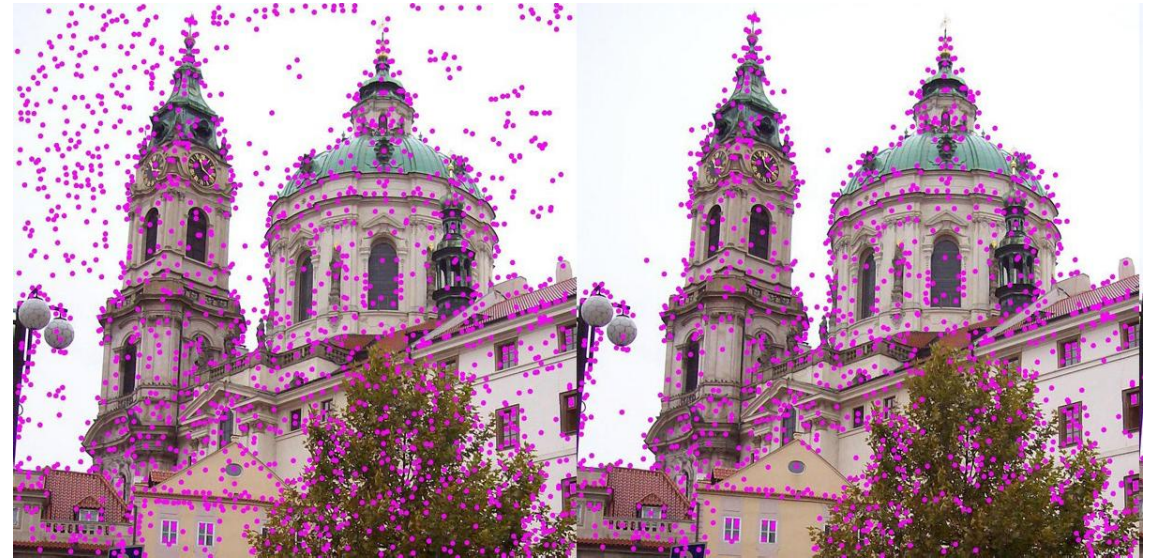


LeCun, Bottou, Bengio and Haffner: Gradient-based learning applied to document recognition. Proceedings of the IEEE, 1998.

1999: SIFT

2000-2010: Feature engineering

- **Scale Invariant** Feature Transform
- Detection and description of salient local features in an image
- Enabled many applications (e.g., image stitching, reconstruction, motion estimation, . . .)



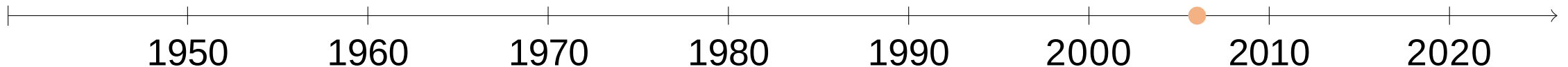
Lowe: Object Recognition from Local Scale-Invariant Features. ICCV, 1999.

2006: Photo Tourism

2000-2010: Feature engineering



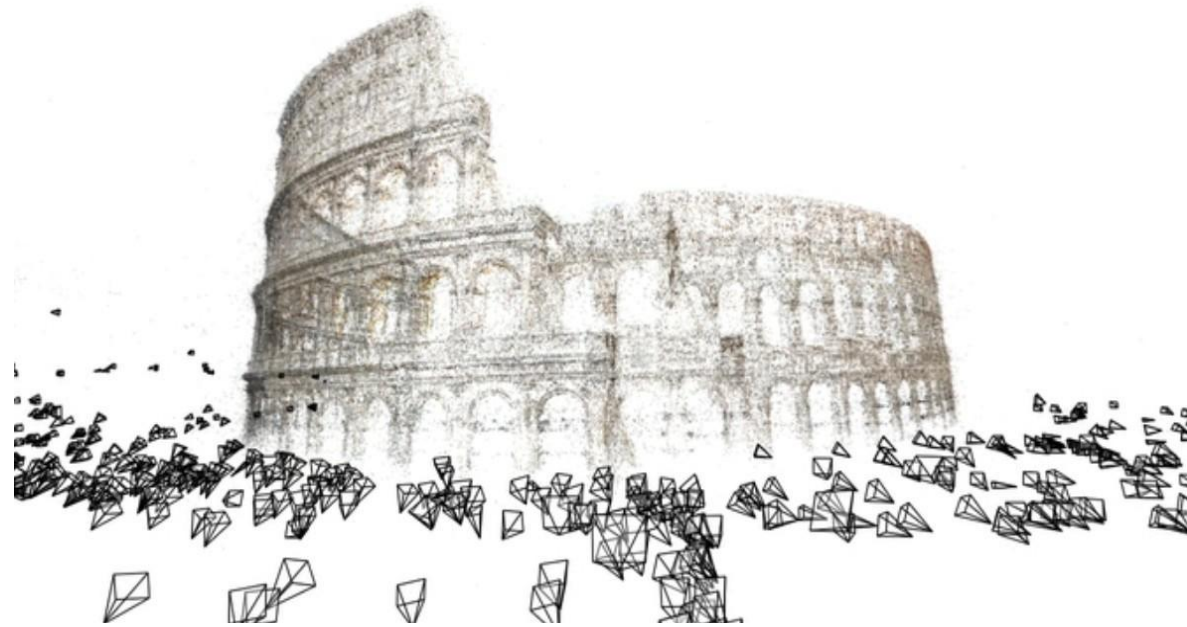
- Large-scale 3D reconstruction from internet photos
- Key ingredients: SIFT feature matching, bundle adjustment



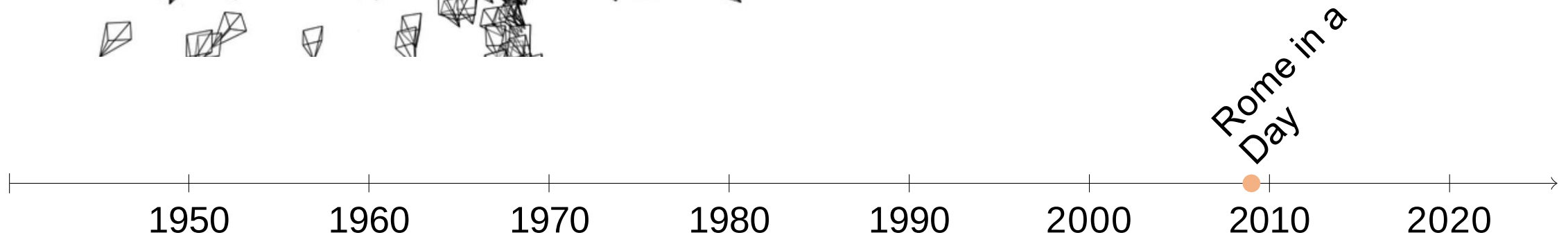
Snavely, Seitz and Szeliski: Photo tourism: exploring photo collections in 3D. SIGGRAPH, 2006.

2009: Building Rome in a Day

2000-2010: Feature engineering



- 3D reconstruction of landmarks and cities from unstructured Internet photo-collections



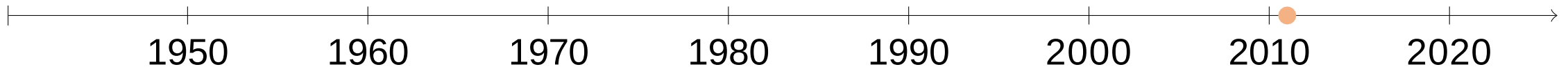
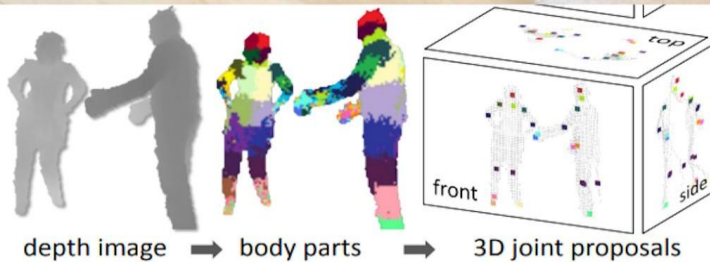
Agarwal, Snavely, Simon, Seitz and Szeliski: Building Rome in a day. ICCV, 2009.

2011: Kinect

2000-2010: Feature engineering



- Active light 3D sensing
- ML for 3D pose estimation
- Multiple hardware generations
- Heavily used for robotics and vision research



Shotton et al.: Real-time human pose recognition in parts from single depth images. CVPR, 2011.

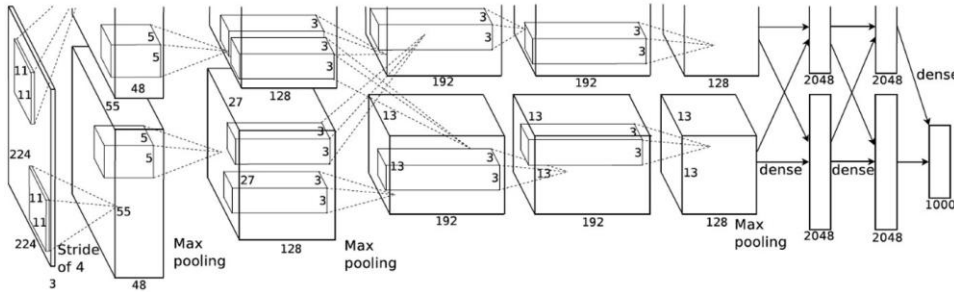
2009-2012: ImageNet and AlexNet

2010-now : Deep learning



ImageNet

- Recognition benchmark (ILSVRC)
- 1 million annotated images in 1k categories



AlexNet

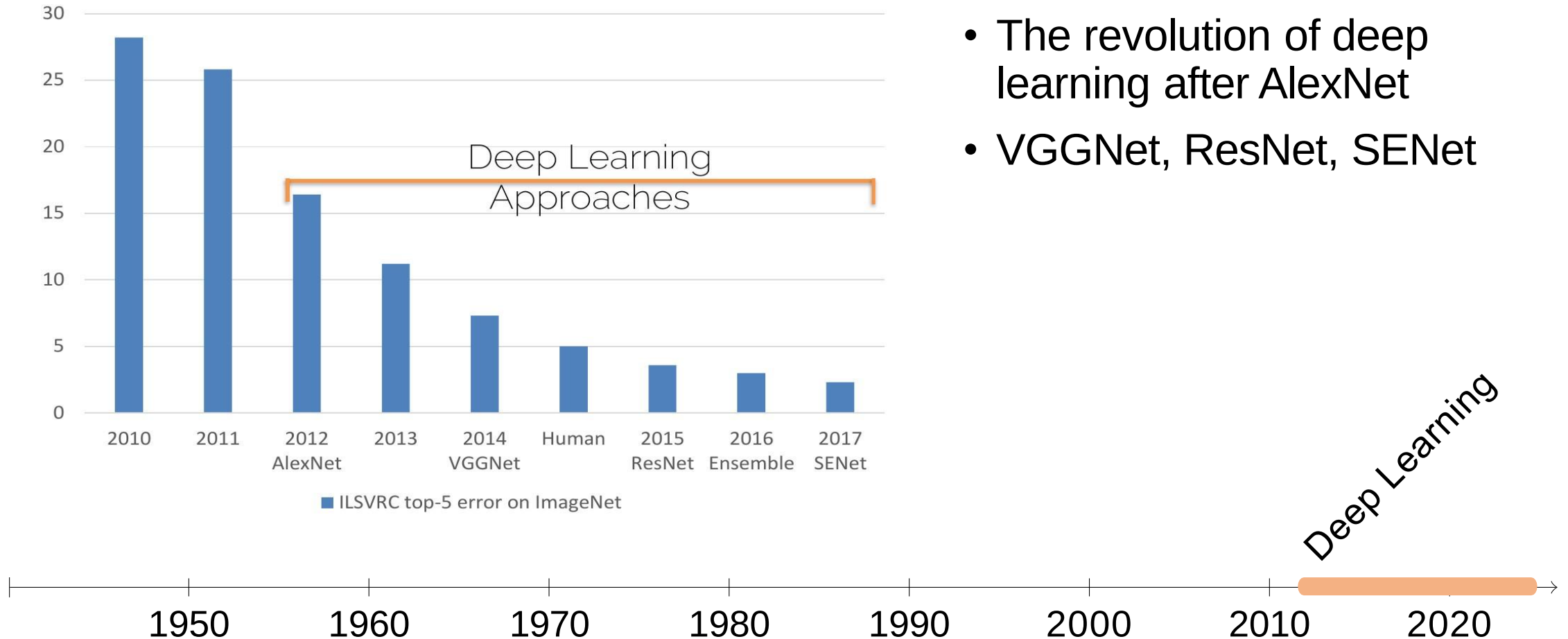
- The first neural network to win ILSVRC via GPU training, deep models, big data



Krizhevsky, Sutskever and Hinton: ImageNet classification with deep convolutional neural networks. NIPS, 2012.

2012 – 2022: Deep Learning Era

2010-now : Deep learning

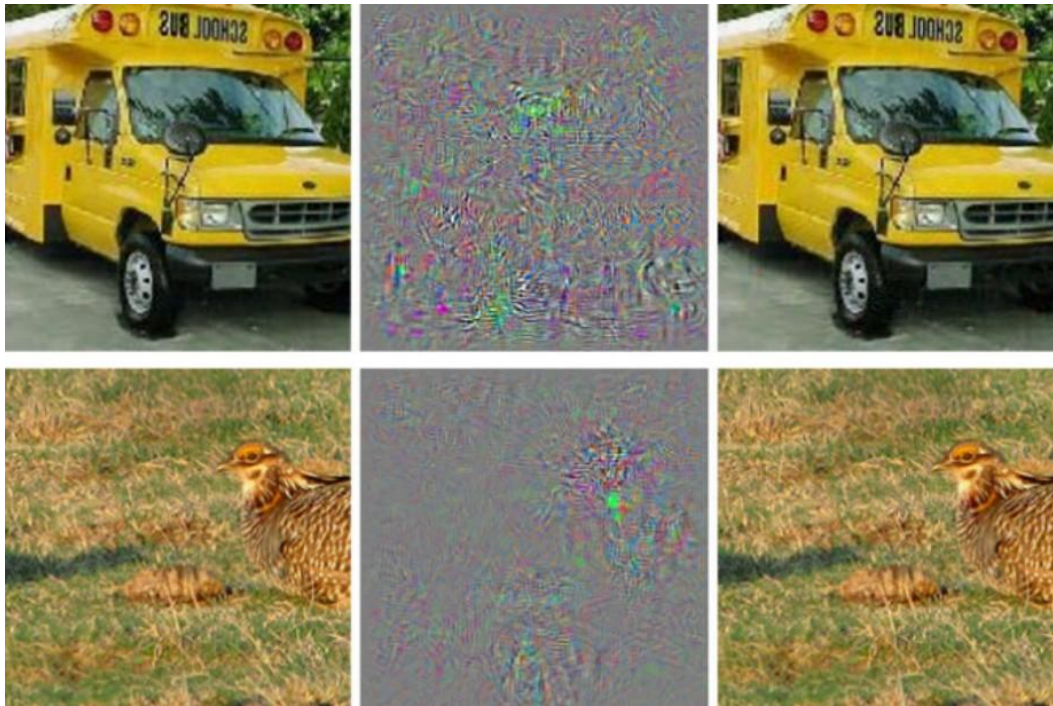


- The revolution of deep learning after AlexNet
- VGGNet, ResNet, SENet

Krizhevsky, Sutskever and Hinton: ImageNet classification with deep convolutional neural networks. NIPS, 2012.

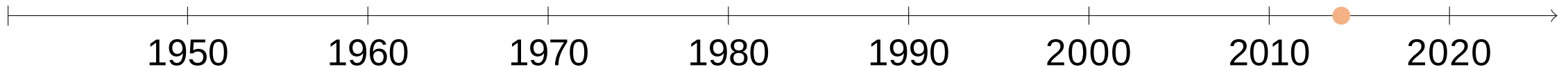
2014: Adversarial Examples

2010-now : Deep learning



- Accurate image classifiers can be fooled by imperceptible changes (here magnified for visibility)
- All images in the right column are classified as “ostrich”

Adv. Examples



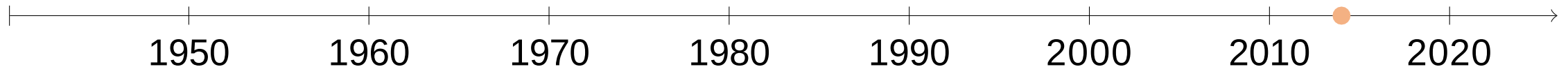
Szegedy et al.: Intriguing properties of neural networks. ICLR, 2014.

2014: Generative Adversarial Networks

2010-now : Deep learning

- Deep generative models (VAEs, GANs) produce compelling images
- StyleGAN2 is state-of-the-art
- Results on faces are hard to distinguish from real images
- Active research on image translation, domain adaptation, content and scene generation and 3D GANs

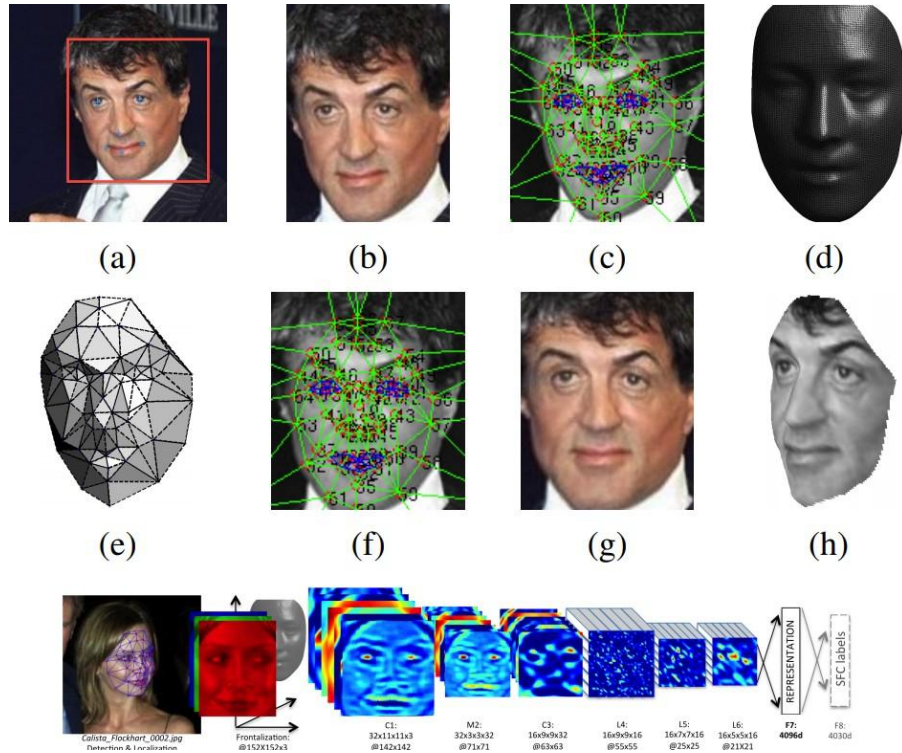
Moore's Law of AI
4.5 years of progress on faces



Goodfellow, Pouget-Abadie, Mirza, Xu, Warde-Farley, Ozair, Courville, Bengio: Generative Adversarial Networks. NIPS, 2014.

2014: DeepFace

2010-now : Deep learning



- Combination of model-based alignment with deep learning for face recognition
- First model to reach human-level face recognition performance

DeepFace



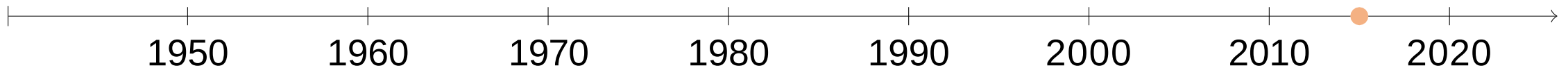
Taigman, et.al. : DeepFace: Closing the Gap to Human-Level Performance in Face Verification. CVPR, 2014.

2015: Deep Reinforcement Learning

2010-now : Deep learning



- Learning a policy (state→action) through random exploration and reward signals (e.g., game score)
- No other supervision
- Success on many Atari games



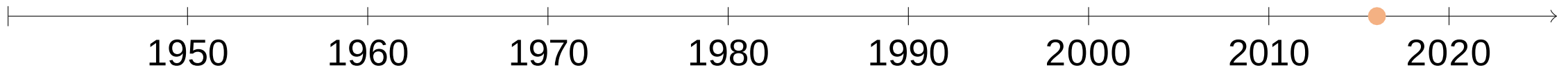
Mnih et al.: Human-level control through deep reinforcement learning. Nature, 2015.

2016: Style Transfer

2010-now : Deep learning



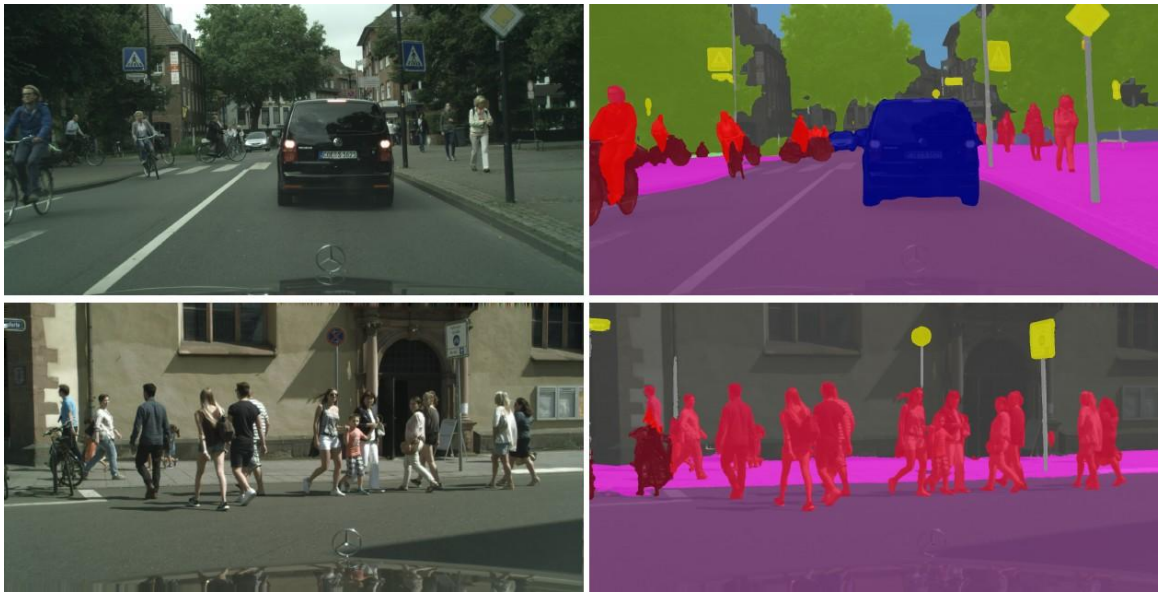
- Manipulate photograph to adopt style of a another image (painting)
- Uses deep network pre-trained on ImageNet for disentangling content from style
- Try yourself: <https://deepart.io>



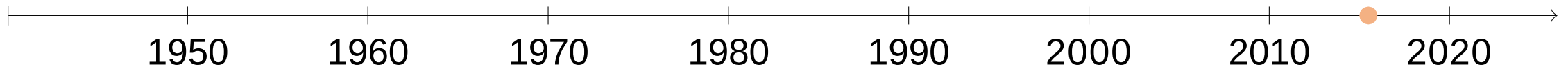
Gatys, Ecker and Bethge: Image Style Transfer Using Convolutional Neural Networks. CVPR, 2016.

2015-2017: Semantic Segmentation

2010-now : Deep learning



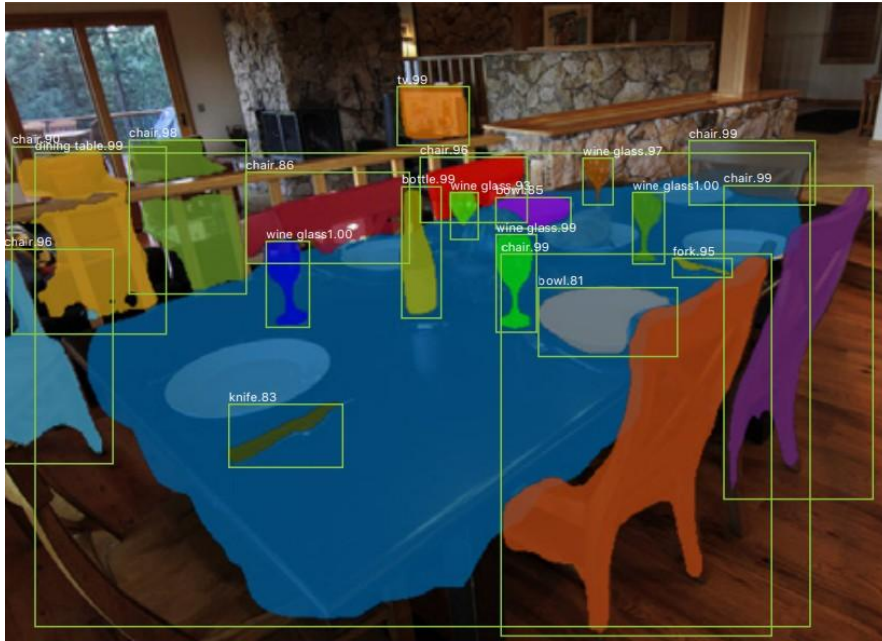
- Assign semantic class to every pixel
- Semantic segmentation starts to work on challenging real-world datasets (e.g., CityScapes)
- 2015: FCN, SegNet
- 2016: DeepLab, FSO
- 2017: DeepLabv3



Kundu, Vineet and Koltun: Feature Space Optimization for Semantic Video Segmentation. CVPR, 2016.

2017: Mask R-CNN

2010-now : Deep learning



- Deep neural network for joint object detection and instance segmentation
- Outputs “structured object”, not only a single number (class label)
- State-of-the-art on MS-COCO

Mask R-CNN



He, Gkioxari, Dollár and Ross Girshick: Mask R-CNN. ICCV, 2017.

2017: Image Captioning

2010-now : Deep learning



"little girl is eating piece of cake."



"baseball player is throwing ball in game."



"woman is holding bunch of bananas."



"black cat is sitting on top of suitcase."



"a young boy is holding a baseball bat."



"a cat is sitting on a couch with a remote control."



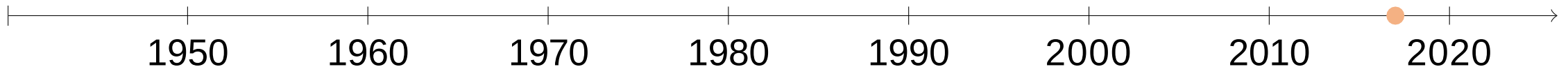
"a woman holding a teddy bear in front of a mirror."



"a horse is standing in the middle of a road."

- Growing interest in combining vision with language
- Several new tasks emerged including image captioning and visual question answering

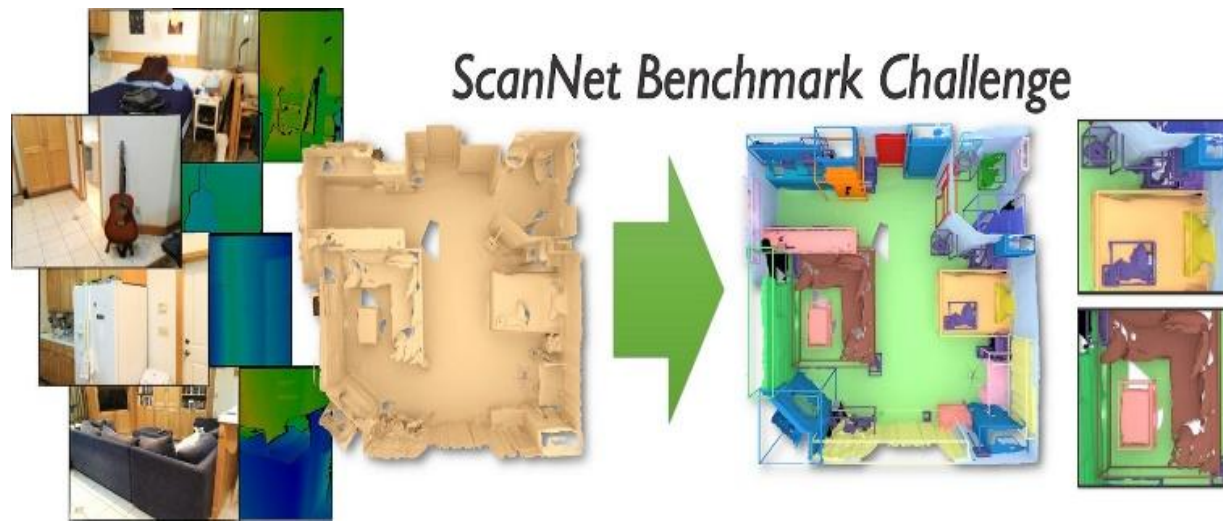
Captioning



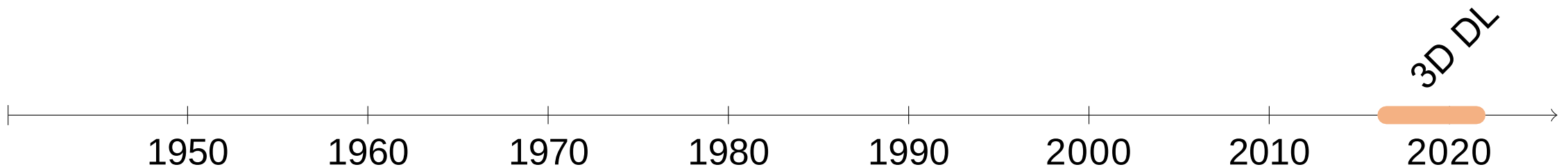
Karpathy and Fei-Fei: Deep Visual-Semantic Alignments for Generating Image Descriptions. PAMI, 2017.

2016-2020: 3D Deep Learning

2010-now : Deep learning



- Voxels, point clouds, meshes, implicit representations
- Prediction of 3D models even from a single image
- Geometry, materials, light



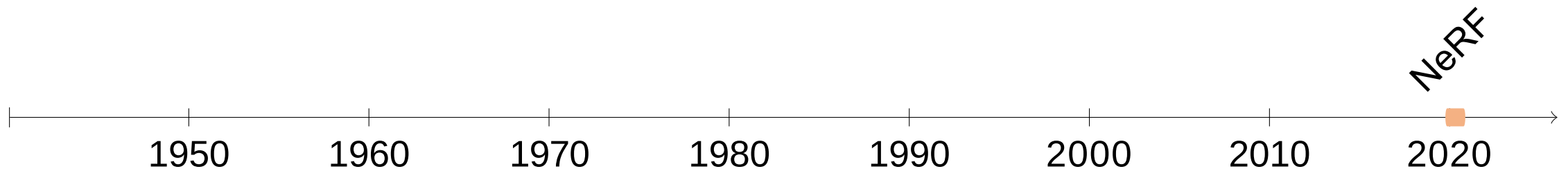
Peng-Shuai Wang et. al.: O-CNN: Octree-based Convolutional Neural Networks for 3D Shape Analysis. SIGGRAPH 2017.

2020: Neural Radiance Field (NeRF)

2010-now : Deep learning



- Synthesizing novel views of complex scenes by optimizing volumetric scene function
- Implicit representation for 3D scene
- Able to encode various geometry, materials, light, motion



Mildenhall, Ben, et al. "Nerf: Representing scenes as neural radiance fields for view synthesis." ECCV 2020

2018-2022: Vision and other modalities

2010-now : Deep learning

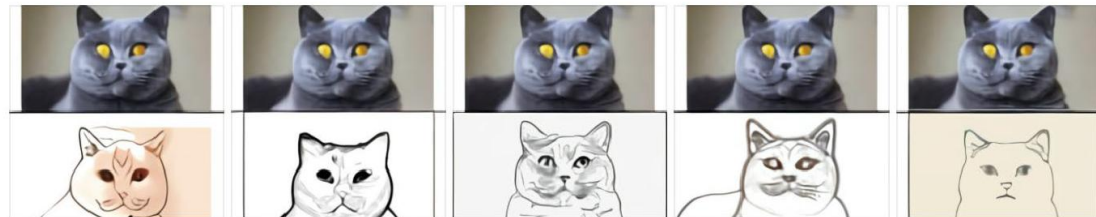
TEXT PROMPT an armchair in the shape of an avocado. . .

AI-GENERATED
IMAGES

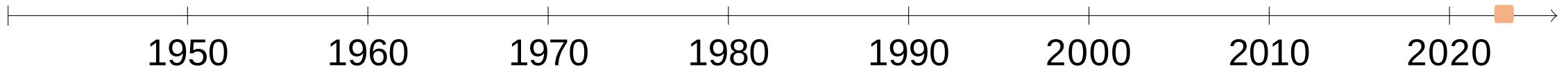


TEXT & IMAGE
PROMPT the exact same cat on the top as a sketch on the bottom

AI-GENERATED
IMAGES



- Generate images from text descriptions, using a dataset of text-image pairs
- Enable zero-shot generation.
- With diverse capabilities, like combining unrelated concepts



Agentic 3D Creation via Joint Agent-Program Design

RUI-HUAN WANG, Peking University, China

SI-TONG WEI, Peking University, China

JIA-QI HE, Peking University, China

HENG-YI WEI, Peking University, China

BAOQUAN CHEN, Peking University, China

PENG-SHUAI WANG*, Peking University, China

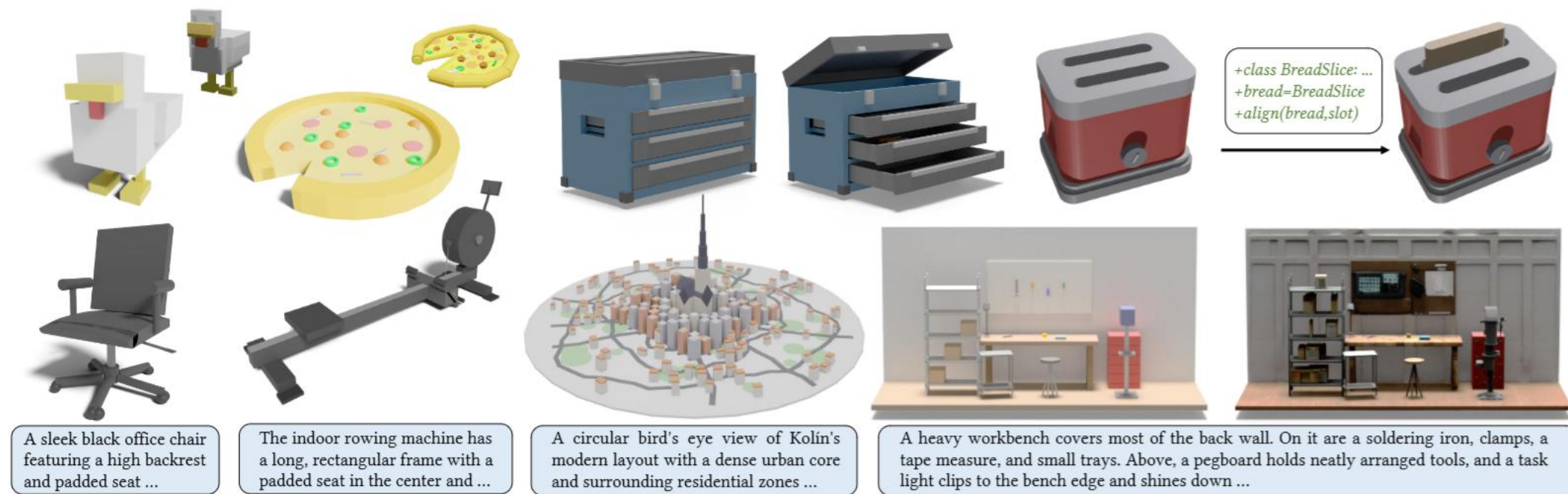


Fig. 1. We present a jointly designed DSL and multi-agent system for robust text- and image-conditioned 3D modeling. The same structured representation also supports downstream applications including articulated asset modeling, shape editing and scene-level creation.



DESIGN BEFORE GEOMETRY

先把想法聊清楚， 再开始构建三维。

这里没有预加载模型，也不会因为普通对话创建资产。你可以讨论需求、形态和结构；明确确认三维生成后，系统才建立新的 GLB 工作会话。

自由讨论与规划

参考图理解

确认后生成 GLB

继续已有 3D 会话

2 个模型



A simple 3D red box with clean edges, matte red material, and balanced proportio

2026/8/17 18:48:07



Create a modern minimalist nightstand/bedside table featuring functional, slidab

2026/8/17 18:47:36



新对话

无 3D session 上下文

Agent ready



ABuilder Agent 页三

你好，我是 ABuilder Agent。我们可以先讨论设计、结构、用途或实现思路；只有当你明确要求创建三维物体并确认生成后，我才会启动 aDSL 管线并建立 3D 会话。

先聊聊你的想法，或明确要创建一个三维物体。



⌚ 普通对话不会创建 session



发送

Agentic Scene Modeling



Create a scene set in a workshop corner.

Components: Interactive Scene Modeling
Workbench, Shelving Unit, Shop Stool, ...

Relations:

- Workbench centered against back wall
- Shelving unit put to the left of workbench
- Shop stool put in front of the workbench
- ...



The scene is already created!



Agentic Scene Modeling



控制



重新加载



Link

名称: KitchenCounter

类型: Visual



按住鼠标左键拖动以控制关节

Summary

- Computer vision
 - Reconstructs meanings (3D, semantics, etc.) from images
 - Image processing; 3D reconstruction; Deep learning
- Computer vision is hard
 - Semantic GAP
 - Projection, viewpoint, deformation, occlusion, illumination, motion, local ambiguities, variations
- History of computer vision
 - 1960-1970: **Heuristics methods**; 1970-1981: **Low-level vision**; 1985-1988: **Neural networks**; 1990-2000: **3D geometry**; 2000-2010: **Feature engineering**; 2010-now : **Deep learning**

